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# Long-staple design approach towards the scalable production of scaffolded DNA origami

Supplementary information

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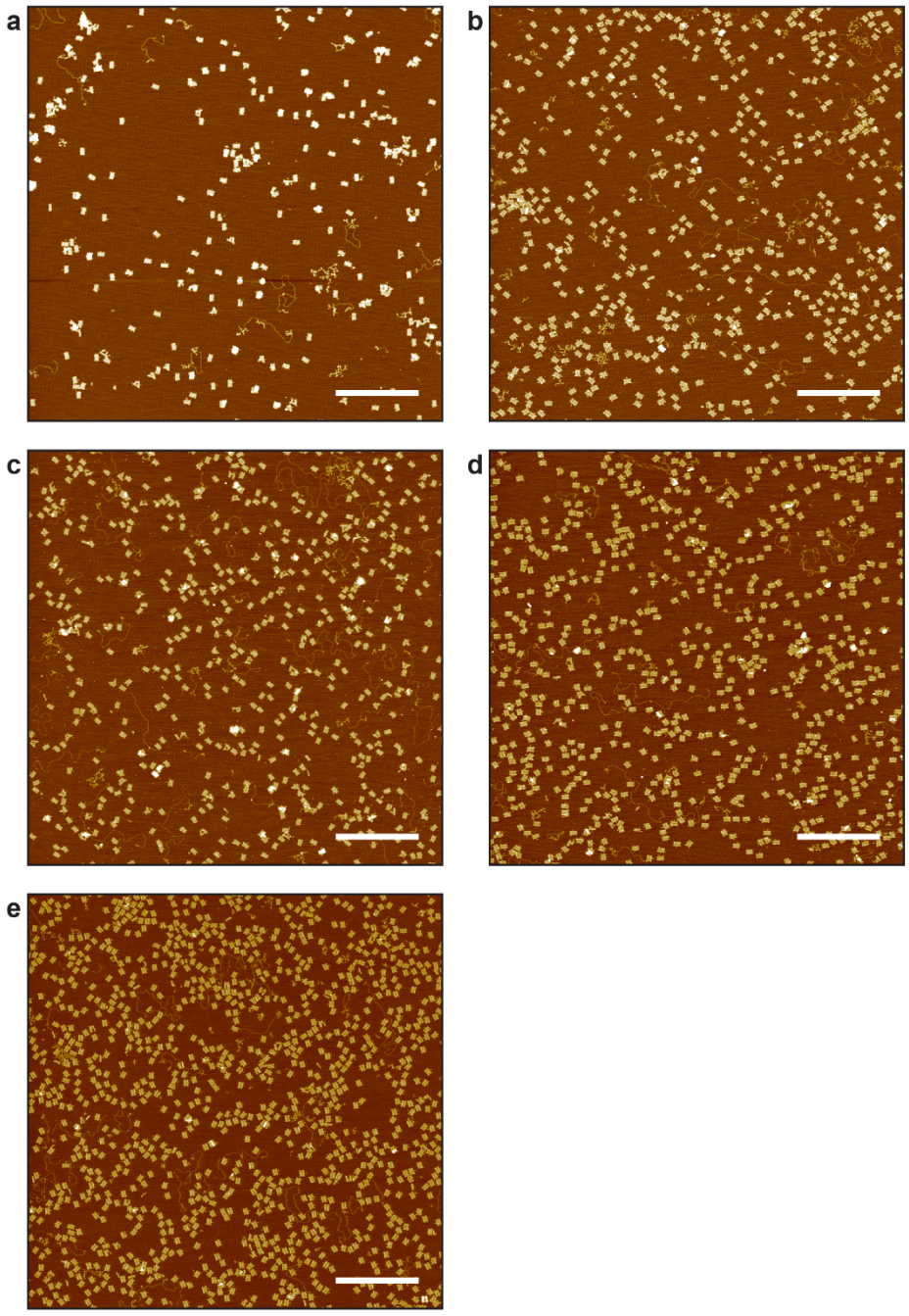



**Figure S1. AFM images of rectangular structures with different staple designs.** (a) 128 nt design with 1:1 scaffold-to-staple ratio. (b) 128 nt design with 1:5 scaffold-to-staple ratio. (c) 192 nt design with 1:1 scaffold-to-staple ratio. (d) 192 nt design with 1:5 scaffold-to-staple ratio. (e) Short staple design. Scale bars: 500 nm.

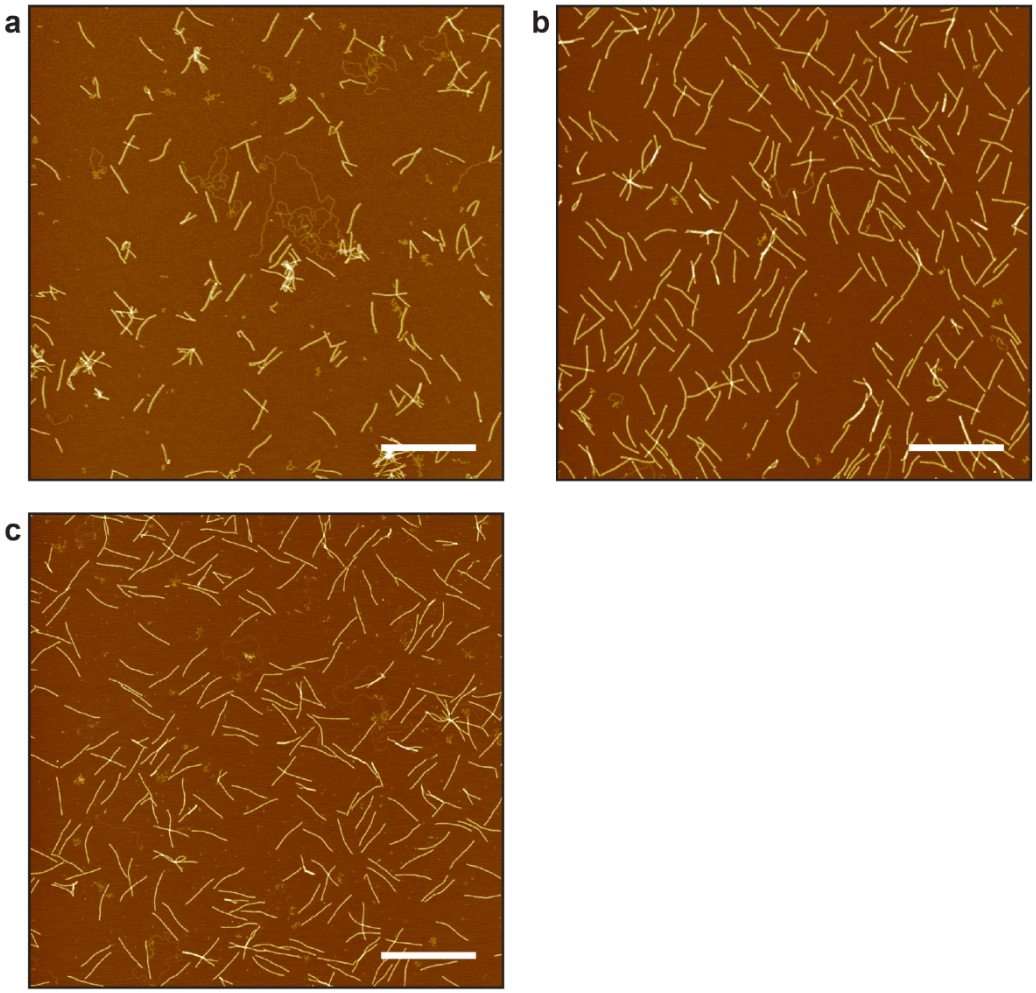



**Figure S2. AFM images of 6HB structures with different staple designs.** (a) 126 nt design with 1:1 scaffold-to-staple ratio. (b) 126 nt design with 1:5 scaffold-to-staple ratio. (c) Short staple design. Scale bars: 500 nm.

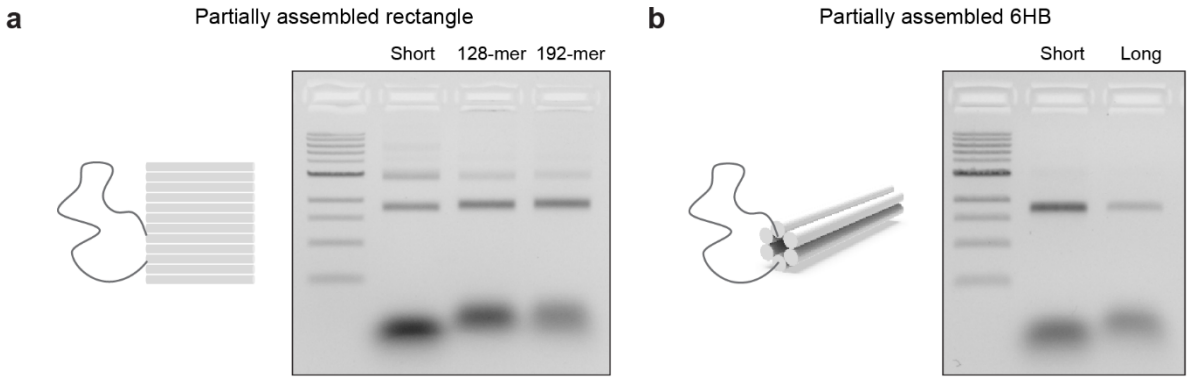



**Figure S3. Comparison of staple length effects using partially assembled structures. (a)** Schematic illustration and agarose gel image of partially formed rectangle structure using only short (32-mer), 128-mer, or 192-mer staple strands. **(b)** Schematic illustration and agarose gel image of partially formed 6HB structures using only short (42-mer) or long (126-mer) staple strands.

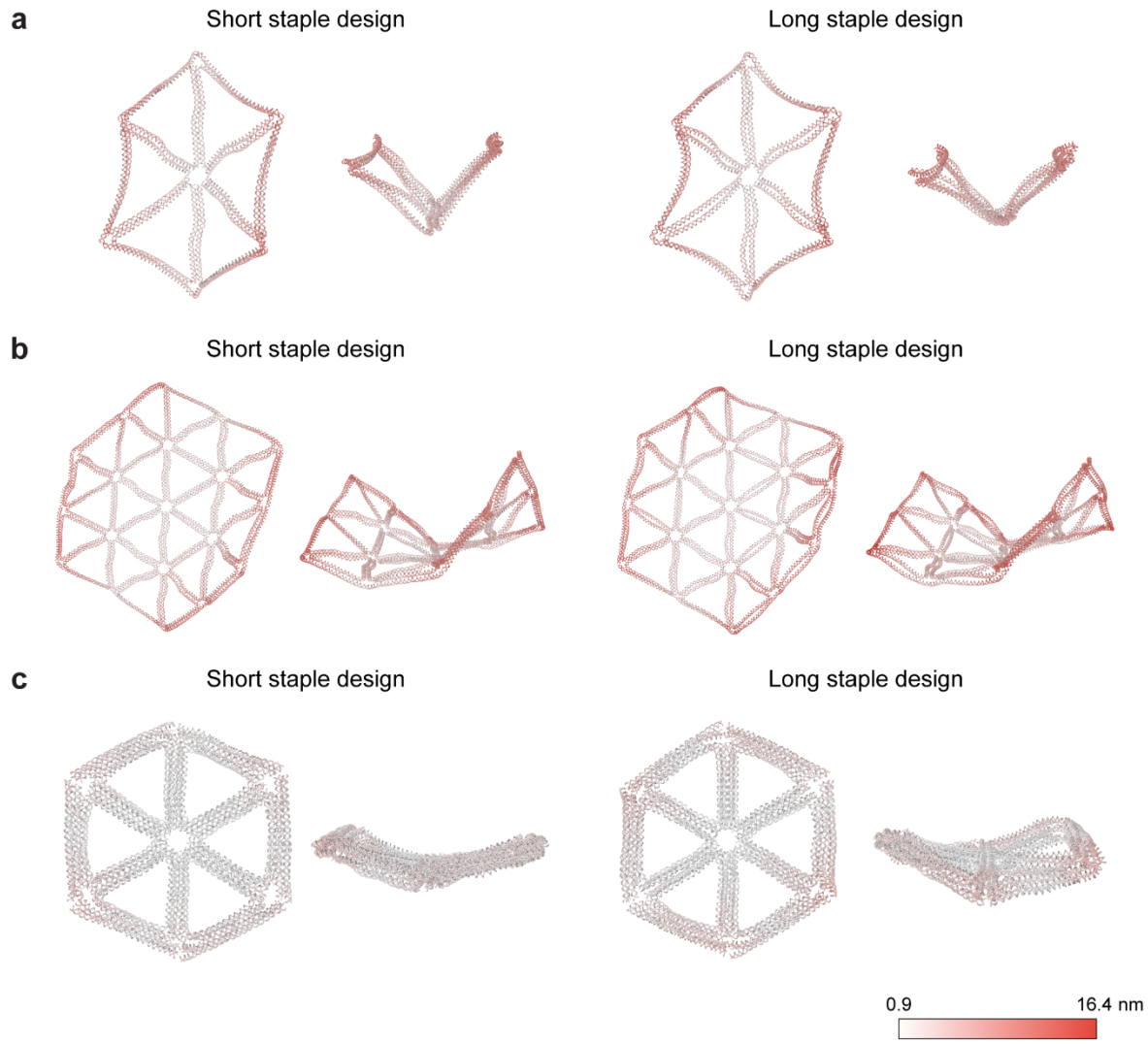



**Figure S4. SNUPI simulation results.** Predicted shapes and root-mean-square fluctuation (RMSF) analyses of the (a) Mini-hexagon, (b) Full-sized hexagon, and (c) Thick-hexagon structures designed with either short or long staples.

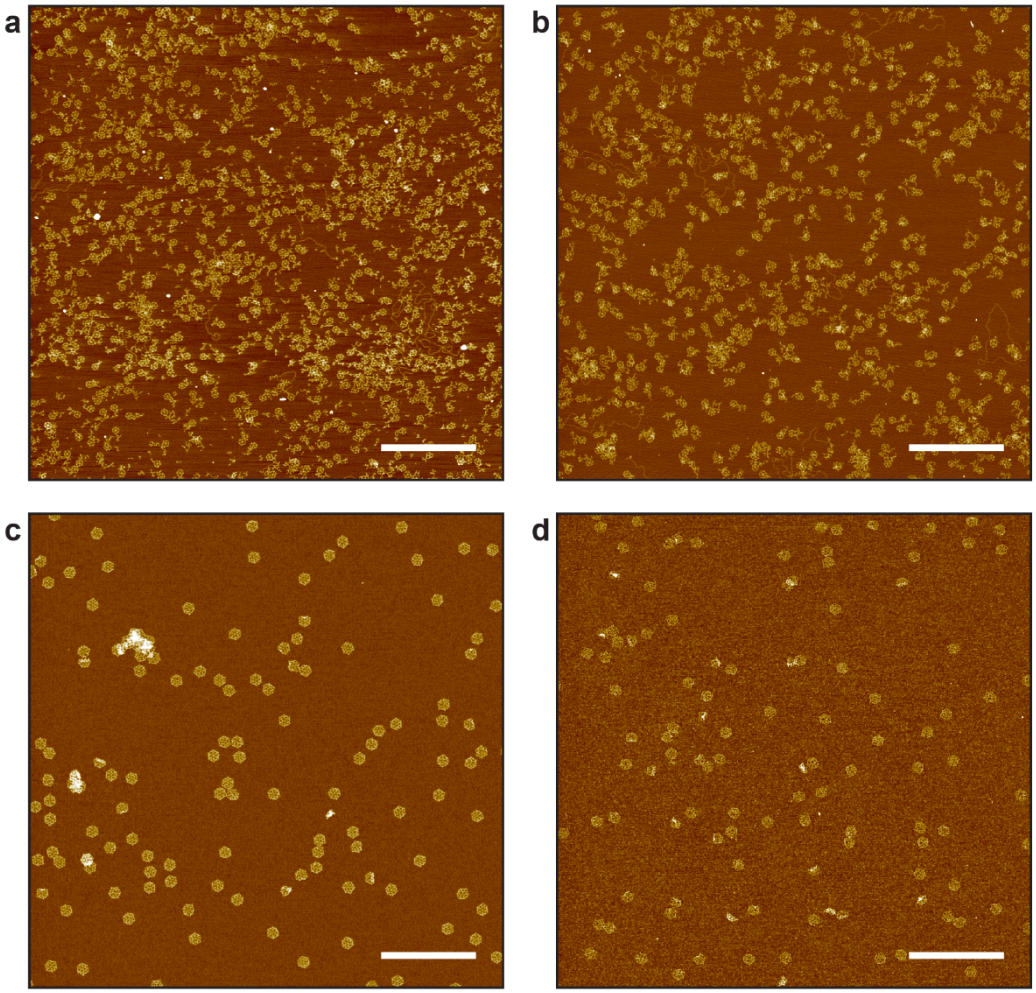



**Figure S5. AFM images of small and full-sized hexagonal structures with different staple designs.** (a) Small hexagon with short staples. (b) Small hexagon with long staples. (c) Full-sized hexagon with short staples. (d) Full-sized hexagon with long staples. Scale bars: 500 nm.

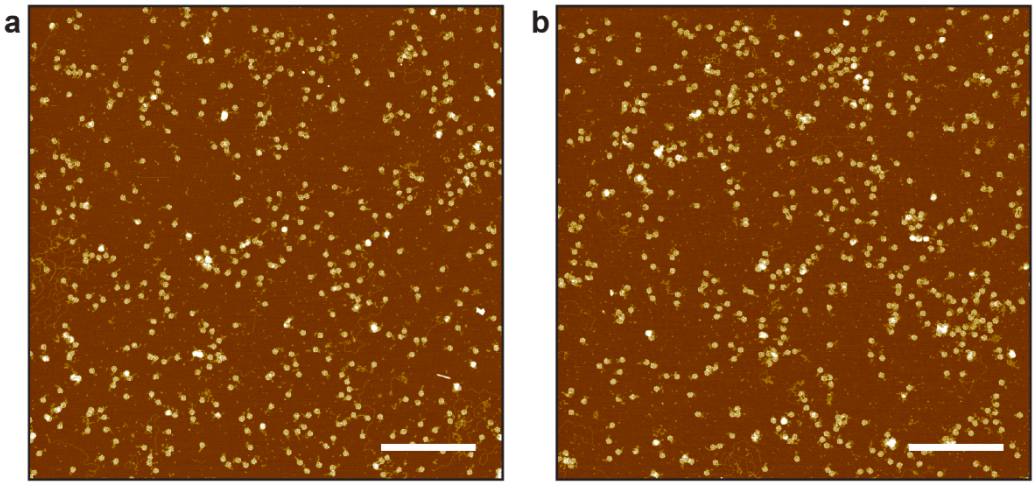



**Figure S6. AFM images of thick hexagonal structures with different staple designs.** (a) Thick hexagon with short staples. (b) Thick hexagon with long staples. Scale bars: 500 nm.

| Staples used for rectangular structure | |
|---|---|
| **Name** | **Sequence (5'→3')** |
| **Rect_B01** | ACGTTAGTAAATGAATTTTCTGTAAGCGGAGT |
| **Rect_B02** | GAGAATAGCTTTTGCGGGATCGTCGGGTAGCA |
| **Rect_B03** | ACGGCTACTTACTTAGCCGGAACGCTGACCAA |
| **Rect_B04** | CTTTGAAAAGAACTGGCTCATTATTTAATAAA |
| **Rect_B05** | ACGAACTAGCGTCCAATACTGCGGAATGCTTT |
| **Rect_B06** | AAACAGTTGATGGCTTAGAGCTTATTTAAATA |
| **Rect_B07** | TGCAACTAAGCAATAAAGCCTCAGTTATGACC |
| **Rect_B08** | CTGTAATATTGCCTGAGAGTCTGGAAAACTAG |
| **Rect_B09** | CATGTCAAGATTCTCCGTGGGAACCGTTGGTG |
| **Rect_B10** | TAGATGGGGGGTAACGCCAGGGTTGTGCCAAG |
| **Rect_B11** | CTTGCATGCATTAATGAATCGGCCCGCCAGGG |
| **Rect_B12** | TGGTTTTTAACGTCAAAGGGCGAAGAACCATC |
| **Rect_C01** | AAAGGCCGAAAGGAACAACTAAAGCTTTCCAG |
| **Rect_C02** | GCTCCATGAGAGGCTTTGAGGACTAGGGAGTT |
| **Rect_C03** | CGATTTTAGAGGACAGATGAACGGCGCGACCT |
| **Rect_C04** | ACTGGATAACGGAACAACATTATTACCTTATG |
| **Rect_C05** | TTTTTGCGCAGAAAACGAGAATGAATGTTTAG |
| **Rect_C06** | CAAAATTAAAGTACGGTGTCTGGAAGAGGTCA |
| **Rect_C07** | TCAGGTCACTTTTGCGGGAGAAGCAGAATTAG |
| **Rect_C08** | ACCCGTCGTCATATGTACCCCGGTAAAGGCTA |
| **Rect_C09** | ATTAAGTTCGCATCGTAACCGTGCGAGTAACA |
| **Rect_C10** | GCCAGCTGCCTGCAGGTCGACTCTGCAAGGCG |
| **Rect_C11** | TGGACTCCCTTTTCACCAGTGAGACCTGTCGT |
| **Rect_C12** | ACCCAAATCAAGTTTTTTGGGGTCAAAGAACG |
| **Rect_D01** | CGTAACGATCTAAAGTTTTGTCGTGAATTGCG |
| **Rect_D02** | AATAATAAGGTCGCTGAGGCTTGCAAAGACTT |
| **Rect_D03** | TTTCATGAAAATTGTGTCGAAATCTGTACAGA |
| **Rect_D04** | CCAGGCGCTTAATCATTGTGAATTACAGGTAG |
| **Rect_D05** | AAAGATTCAGGGGGTAATAGTAAACCATAAAT |
| **Rect_D06** | CAAAAATCATTGCTCCTTTTGATAAGTTTCAT |
| **Rect_D07** | TCCATATACATACAGGCAAGGCAACTTTATTT |
| **Rect_D08** | CAACGCAATTTTTGAGAGATCTACTGATAATC |
| **Rect_D09** | AGAAAAGCAACATTAAATGTGAGCATCTGCCA |
| **Rect_D10** | GTTTGAGGGAAAGGGGGATGTGCTAGAGGATC |
| **Rect_D11** | CCCGGGTACTTTCCAGTCGGGAAACGGGCAAC |
| **Rect_D12** | AGCTGATTACAAGAGTCCACTATTGAGGTGCC |
| **Rect_E01** | ATATATTCTTTTTTCACGTTGAAAATAGTTAG |
| **Rect_E02** | CGCCTGATGGAAGTTTCCATTAAACATAACCG |
| **Rect_E03** | TTTCAACTATAGGCTGGCTGACCTTGTATCAT |
| **Rect_E04** | TTTGCCAGATCAGTTGAGATTTAGTGGTTTAA |
| **Rect_E05** | TACCTTTAAGGTCTTTACCCTGACAAAGAAGT |
| **Rect_E06** | CAATAAATACAGTTGATTCCCAATTTAGAGAG |
| **Rect_E07** | GGTAGCTAGGATAAAAATTTTTAGTTAACATC |
| **Rect_E08** | CTTTCATCCCCAAAAACAGGAAGACCGGAGAG |
| **Rect_E09** | CAGCTGGCGGACGACGACAGTATCGTAGCCAG |
| **Rect_E10** | ACTGCCCGCCGAGCTCGAATTCGTTATTACGC |
| **Rect_E11** | AGTTTGGAGCCCTTCACCGCCTGGTTGCGCTC |
| **Rect_E12** | GTAAAGCACTAAATCGGAACCCTAGTTGTTCC |
| **Rect_F01** | TGTAGCATTCCACAGACAGCCCTCATCTCCAA |
| **Rect_F02** | AAAAAAGGACAACCATCGCCCACGCGGGTAAA |
| **Rect_F03** | ATACGTAAAAGTACAACGGAGATTTCATCAAG |
| **Rect_F04** | AGTAATCTTAAATTGGGCTTGAGAGAATACCA |
| **Rect_F05** | CATTCAACGCGAGAGGCTTTTGCATATTATAG |
| **Rect_F06** | TCAGAAGCCTCCAACAGGTCAGGATCTGCGAA |
| **Rect_F07** | CGAGTAGAACTAATAGTAGTAGCAAACCCTCA |
| **Rect_F08** | TATATTTTAGCTGATAAATTAATGTTGTATAA |
| **Rect_F09** | GCAAATATCGCGTCTGGCCTTCCTGGCCTCAG |
| **Rect_F10** | GAAGATCGGTGCGGGCCTCTTCGCAATCATGG |
| **Rect_F11** | TCATAGCTACTCACATTAATTGCGCCCTGAGA |
| **Rect_F12** | GAGTTGCACGAGATAGGGTTGAGTAAGGGAGC |
| **Rect_G01** | CAATGACACTCCAAAAGGAGCCTTACAACGCC |
| **Rect_G02** | GCGAAACATGCCACTACGAAGGCATGCGCCGA |
| **Rect_G03** | ACGAGTAGTGACAAGAACCGGATATACCAAGC |
| **Rect_G04** | CCAAAATATAATGCAGATACATAAACACCAGA |

| | |
|---|---|
| **Rect_G05** | GAAGCAAAAAAGCGGATTGCATCAGATAAAAA |
| **Rect_G06** | TCAATTCTTTTAGTTTGACCATTACCAGACCG |
| **Rect_G07** | ACCGTTCTAAATGCAATGCCTGAGAGGTGGCA |
| **Rect_G08** | AAATAATTTTAAATTGTAAACGTTGATATTCA |
| **Rect_G09** | GGCGATCGCACTCCAGCCAGCTTTGCCATCAA |
| **Rect_G10** | GTGAGCTAGTTTCCTGTGTGAAATTTGGGAAG |
| **Rect_G11** | GAATAGCCGCAAGCGGTCCACGCTCCTAATGA |
| **Rect_G12** | CCCCGATTTAGAGCTTGACGGGGAAATCAAAA |
| **Rect_H01** | TGAGTTTCGTCACCAGTACAAACTTAATTGTA |
| **Rect_H02** | TCGGTTTAGCTTGATACCGATAGTCCAACCTA |
| **Rect_H03** | AAACGAAATGACCCCCAGCGATTATTCATTAC |
| **Rect_H04** | CCAAATCACTTGCCCTGACGAGAACGCCAAAA |
| **Rect_H05** | GGAATTACTCGTTTACCAGACGACAAAAGATT |
| **Rect_H06** | AAGAGGAACGAGCTTCAAAGCGAAGATACATT |
| **Rect_H07** | TCGCAAATGGGGCGCGAGCTGAAATAATGTGT |
| **Rect_H08** | AGGTAAAGAAATCACCATCAATATAATATTTT |
| **Rect_H09** | GTTAAAATTTTAACCAATAGGAACCCGGCACC |
| **Rect_H10** | GCTTCTGGTCAGGCTGCGCAACTGTGTTATCC |
| **Rect_H11** | GCTCACAATGTAAAGCCTGGGGTGGGTTTGCC |
| **Rect_H12** | CCAGCAGGGGCAAAATCCCTTATAAAGCCGGC |
| **Rect_I01** | CTTAAACATCAGCTTGCTTTCGAGAAACAGTT |
| **Rect_I02** | CTCATCTTGAGGCAAAAGAATACACTCCCTCA |
| **Rect_I03** | GAATAAGGACGTAACAAAGCTGCTGACGGAAA |
| **Rect_I04** | CATAACCCGAGGCATAGTAAGAGCTTTTTAAG |
| **Rect_I05** | TTTTAATTGCCCGAAAGACTTCAATTCCAGAG |
| **Rect_I06** | TTTCATTTGGTCAATAACCTGTTTAATCAATA |
| **Rect_I07** | AGACAGTCATTCAAAAGGGTGAGATATCATAT |
| **Rect_I08** | GCTCATTTTCGCATTAAATTTTTGAGCTTAGA |
| **Rect_I09** | TTCGCCATTGCCGGAAACCAGGCAAACAGTAC |
| **Rect_I10** | GCATAAAGTTCCACACAACATACGAAACAATT |
| **Rect_I11** | CCGAAATCCGAAAATCCTGTTTGAAATACCGA |
| **Rect_I12** | GAACGTGGCGAGAAAGGAAGGGAACAAACTAT |
| **Rect_J01** | CAAGCCCAATAGGAACCCATGTACCGTAACAC |
| **Rect_J02** | AATGCCCCGTAACAGTGCCCGTATGTGAATTT |
| **Rect_J03** | GAGCCGCCCCACCACCGGAACCGCCTAAAACA |
| **Rect_J04** | TTATTCATAGGGAAGGTAAATATTCATTCAGT |
| **Rect_J05** | AAAAGTAATATCTTACCGAAGCCCAACACTAT |
| **Rect_J06** | CCTAATTTACGCTAACGAGCGTCTATATCGCG |
| **Rect_J07** | ATCGGCTGCGAGCATGTAGAAACCAGCTATAT |
| **Rect_J08** | GCGTTATAGAAAAAGCCTGTTTAGAAGGCCGG |
| **Rect_J09** | TTAAGACGTTGAAAACATAGCGATTTAAATCA |
| **Rect_J10** | CTTTTACACAGATGAATATACAGTAAGCGCCA |
| **Rect_J11** | CGACAACTAAGTATTAGACTTTACAGCCGGAA |
| **Rect_J12** | ACGAACCAAAACATCGCCATTAAATGGTGGTT |
| **Rect_K01** | TGCCTTGACTGCCTATTTCGGAACAGGGATAG |
| **Rect_K02** | AACCAGAGACCCTCAGAACCGCCAGGGGTCAG |
| **Rect_K03** | ATTGAGGGTAAAGGTGAATTATCAATCACCGG |
| **Rect_K04** | GCAATAGCGCAGATAGCCGAACAATTCAACCG |
| **Rect_K05** | TCTTACCAGCCAGTTACAAAATAAATGAAATA |
| **Rect_K06** | CTAATTTATCTTTCCTTATCATTCATCCTGAA |
| **Rect_K07** | AATTACTACAAATTCTTACCAGTAATCCCATC |
| **Rect_K08** | TAGAATCCCTGAGAAGAGTCAATAGGAATCAT |
| **Rect_K09** | TTTAACGTTCGGGAGAAACAATAATTTTCCCT |
| **Rect_K10** | GGATTTAGCGTATTAAATCCTTTGTTTTCAGG |
| **Rect_K11** | TAGCCCTACCAGCAGAAGATAAAAACATTTGA |
| **Rect_K12** | CGGCCTTGCTGGTAATATCCAGAACGAACTGA |
| **Rect_L01** | CTCAGAGCCACCACCCTCATTTTCCTATTATT |
| **Rect_L02** | CTGAAACAGGTAATAAGTTTTAACCCCTCAGA |
| **Rect_L03** | GCCACCACTCTTTTCATAATCAAACCGTCACC |
| **Rect_L04** | GACTTGAGAGACAAAAGGGCGACAAGTTACCA |
| **Rect_L05** | GAAGGAAAATAAGAGCAAGAAACAACAGCCAT |
| **Rect_L06** | ATTATTTAACCCAGCTACAATTTTCAAGAACG |
| **Rect_L07** | GGTATTAAGAACAAGAAAAATAATTAAAGCCA |
| **Rect_L08** | ACGCTCAAAATAAGAATAAACACCGTGAATTT |
| **Rect_L09** | ATCAAAATCGTCGCTATTAATTAACGGATTCG |
| **Rect_L10** | CCTGATTGAAAGAAATTGCGTAGACCCGAACG |

| | |
|---|---|
| **Rect_L11** | TTATTAATGCCGTCAATAGATAATCAGAGGTG |
| **Rect_L12** | AGGCGGTCATTAGTCTTTAATGCGCAATATTA |
| **Rect_M01** | AGTGTACTTGAAAGTATTAAGAGGCCGCCACC |
| **Rect_M02** | GTTTGCCACCTCAGAGCCGCCACCGATACAGG |
| **Rect_M03** | AGCGCCAACCATTTGGGAATTAGATTATTAGC |
| **Rect_M04** | GCCCAATACCGAGGAAACGCAATAGGTTTACC |
| **Rect_M05** | TATTTTGCTCCCAATCCAAATAAGTGAGTTAA |
| **Rect_M06** | TAAGTCCTACCAAGTACCGCACTCTTAGTTGC |
| **Rect_M07** | AGGCGTTACAGTAGGGCTTAATTGACAATAGA |
| **Rect_M08** | CTGTAAATCATAGGTCTGAGAGACGATAAATA |
| **Rect_M09** | ACAGAAATCTTTGAATACCAAGTTCCTTGCTT |
| **Rect_M10** | AGATTAGATTTAAAAGTTTGAGTACACGTAAA |
| **Rect_M11** | GAATGGCTAGTATTAACACCGCCTCAACTAAT |
| **Rect_M12** | CCGCCAGCCATTGCAACAGGAAAAATATTTTT |
| **Rect_N01** | CCCTCAGAACCGCCACCCTCAGAACTGAGACT |
| **Rect_N02** | CCTCAAGAATACATGGCTTTTGATAGAACCAC |
| **Rect_N03** | CACCAGAGTTCGGTCATAGCCCCCGCCAGCAA |
| **Rect_N04** | AATCACCAAATAGAAAATTCATATATAACGGA |
| **Rect_N05** | ATACCCAAGATAACCCACAAGAATAAACGATT |
| **Rect_N06** | TTTTGTTTAAGCCTTAAATCAAGAATCGAGAA |
| **Rect_N07** | CAAGCAAGACGCGCCTGTTTATCAAGAATCGC |
| **Rect_N08** | CATATTTAGAAATACCGACCGTGTTACCTTTT |
| **Rect_N09** | TAACCTCCATATGTGAGTGAATAAACAAAATC |
| **Rect_N10** | GCGCAGAGATATCAAAATTATTTGACATTATC |
| **Rect_N11** | ATTTTGCGTCTTTAGGAGCACTAAGCAACAGT |
| **Rect_N12** | GCCACGCTATACGTGGCACAGACAACGCTCAT |
| **Rect_O01** | TAAGCGTCGAAGGATTAGGATTAGTACCGCCA |
| **Rect_O02** | TCGGCATTCCGCCGCCAGCATTGACGTTCCAG |
| **Rect_O03** | TCACAATCGTAGCACCATTACCATCGTTTTCA |
| **Rect_O04** | ATCAGAGAAAGAACTGGCATGATTTTATTTTG |
| **Rect_O05** | AGGTTTTGAACGTCAAAAATGAAAGCGCTAAT |
| **Rect_O06** | AATGCAGACCGTTTTTATTTTCATCTTGCGGG |
| **Rect_O07** | AATGGTTTACAACGCCAACATGTAGTTCAGCT |
| **Rect_O08** | AAATCAATGGCTTAGGTTGGGTTACTAAATTT |
| **Rect_O09** | AACCTACCGCGAATTATTCATTTCCAGTACAT |
| **Rect_O10** | CTAAAATAGAACAAAGAAACCACCAGGGTTAG |
| **Rect_O11** | GCGTAAGAGAGAGCCAGCAGCAAAAAGGTTAT |
| **Rect_O12** | GGAAATACCTACATTTTGACGCTCACCTGAAA |
| **Rect_P01** | TATCACCGTACTCAGGAGGTTTAGCGGGGTTT |
| **Rect_P02** | TGCTCAGTCAGTCTCTGAATTTACCAGGAGGT |
| **Rect_P03** | TGAGGCAGGCGTCAGACTGTAGCGTAGCAAGG |
| **Rect_P04** | CCGGAAACACACCACGGAATAAGTAAGACTCC |
| **Rect_P05** | TTATTACGGTCAGAGGGTAATTGAATAGCAGC |
| **Rect_P06** | CTTTACAGTTAGCGAACCTCCCGACGTAGGAA |
| **Rect_P07** | TCATTACCCGACAATAAACAACATATTTAGGC |
| **Rect_P08** | AGAGGCATAATTTCATCTTCTGACTATAACTA |
| **Rect_P09** | TATGTAAACCTTTTTTAATGGAAAAATTACCT |
| **Rect_P10** | GAGCAAAAACTTCTGAATAATGGAAGAAGGAG |
| **Rect_P11** | CGGAATTATTGAAAGGAATTGAGGTGAAAAAT |
| **Rect_P12** | CTAAAGCAAGATAGAACCCTTCTGAATCGTCT |
| **Rect_Q01** | GGAAAGCGACCAGGCGGATAAGTGAATAGGTG |
| **Rect_Q02** | TGCCTTTAGTCAGACGATTGGCCTGCCAGAAT |
| **Rect_Q03** | ACGCAAAGGTCACCAATGAAACCAATCAAGTT |
| **Rect_Q04** | TGAACAAACAGTATGTTAGCAAACTAAAAGAA |
| **Rect_Q05** | GAGGCGTTAGAGAATAACATAAAAGAACACCC |
| **Rect_Q06** | CCAGACGAGCGCCCAATAGCAAGCAAGAACGC |
| **Rect_Q07** | TTTTAGTTTTTCGAGCCAGTAATAAATTCTGT |
| **Rect_Q08** | TTGAATTATGCTGATGCAAATCCACAAATATA |
| **Rect_Q09** | TGGATTATGAAGATGATGAAACAAAATTTCAT |
| **Rect_Q10** | ATCAACAGTCATCATATTCCTGATTGATTGTT |
| **Rect_Q11** | GCCAACAGTCACCTTGCTGAACCTGTTGGCAA |
| **Rect_Q12** | GAAATGGATTATTTACATTGGCAGACATTCTG |
| **Rect_128nt_B01** | ACGTTAGTAAATGAATTTTCTGTAAGCGGAGTGAGAATAGCTTTTGCGGGATCGTCGGGTAGCAACGGCTACTTACTTAG<br>CCGGAACGCTGACCAACTTTGAAAAGAACTGGCTCATTATTTAATAAA |
| **Rect_128nt_B02** | ACGAACTAGCGTCCAATACTGCGGAATGCTTTAAACAGTTGATGGCTTAGAGCTTATTTAAATATGCAACTAAGCAATAA<br>AGCCTCAGTTATGACCCTGTAATATTGCCTGAGAGTCTGGAAAACTAG |
| **Rect_128nt_B03** | CATGTCAAGATTCTCCGTGGGAACCGTTGGTGTAGATGGGGGGTAACGCCAGGGTTGTGCCAAGCTTGCATGCATTAATG |

| | |
|---|---|
| | AATCGGCCCGCCAGGGTGGTTTTTAACGTCAAAGGGCGAAGAACCATC |
| **Rect_128nt_C01** | ACTGGATAACGGAACAACATTATTACCTTATGCGATTTTAGAGGACAGATGAACGGCGCGACCTGCTCCATGAGAGGCTT TGAGGACTAGGGAGTTAAAGGCCGAAAGGAACAACTAAAGCTTTCCAG |
| **Rect_128nt_C02** | ACCCGTCGTCATATGTACCCCGGTAAAGGCTATCAGGTCACTTTTGCGGGAGAAGCAGAATTAGCAAAATTAAAGTACGG TGTCTGGAAGAGGTCATTTTTGCGCAGAAAACGAGAATGAATGTTTAG |
| **Rect_128nt_C03** | ACCCAAATCAAGTTTTTTGGGGTCAAAGAACGTGGACTCCCTTTTCACCAGTGAGACCTGTCGTGCCAGCTGCCTGCAGG TCGACTCTGCAAGGCGATTAAGTTCGCATCGTAACCGTGCGAGTAACA |
| **Rect_128nt_D01** | CGTAACGATCTAAAGTTTTGTCGTGAATTGCGAATAATAAGGTCGCTGAGGCTTGCAAAGACTTTTTCATGAAAATTGTG TCGAAATCTGTACAGACCAGGCGCTTAATCATTGTGAATTACAGGTAG |
| **Rect_128nt_D02** | AAAGATTCAGGGGGTAATAGTAAACCATAAATCAAAAATCATTGCTCCTTTTGATAAGTTTCATTCCATATACATACAGG CAAGGCAACTTTATTTCAACGCAATTTTTGAGAGATCTACTGATAATC |
| **Rect_128nt_D03** | AGAAAAGCAACATTAAATGTGAGCATCTGCCAGTTTGAGGGAAAGGGGGATGTGCTAGAGGATCCCCGGGTACTTTCCA GTCGGGAAACGGGCAACAGCTGATTACAAGAGTCCACTATTGAGGTGCC |
| **Rect_128nt_E01** | TTTGCCAGATCAGTTGAGATTTAGTGGTTTAATTTCAACTATAGGCTGGCTGACCTTGTATCATCGCCTGATGGAAGTTTC CATTAAACATAACCGATATATTCTTTTTTCACGTTGAAAATAGTTAG |
| **Rect_128nt_E02** | CTTTCATCCCCAAAAACAGGAAGACCGGAGAGGGTAGCTAGGATAAAAATTTTTAGTTAACATCCAATAAATACAGTTG ATTCCCAATTTAGAGAGTACCTTTAAGGTCTTTACCCTGACAAAGAAGT |
| **Rect_128nt_E03** | GTAAAGCACTAAATCGGAACCCTAGTTGTTCCAGTTTGGAGCCCTTCACCGCCTGGTTGCGCTCACTGCCCGCCGAGCTC GAATTCGTTATTACGCCAGCTGGCGGACGACGACAGTATCGTAGCCAG |
| **Rect_128nt_F01** | TGTAGCATTCCACAGACAGCCCTCATCTCCAAAAAAAAGGACAACCATCGCCCACGCGGGTAAAATACGTAAAAGTACA ACGGAGATTTCATCAAGAGTAATCTTAAATTGGGCTTGAGAGAATACCA |
| **Rect_128nt_F02** | CATTCAACGCGAGAGGCTTTTGCATATTATAGTCAGAAGCCTCCAACAGGTCAGGATCTGCGAACGAGTAGAACTAATAG TAGTAGCAAACCCTCATATATTTTAGCTGATAAATTAATGTTGTATAA |
| **Rect_128nt_F03** | GCAAATATCGCGTCTGGCCTTCCTGGCCTCAGGAAGATCGGTGCGGGCCTCTTCGCAATCATGGTCATAGCTACTCACATT AATTGCGCCCTGAGAGAGTTGCACGAGATAGGGTTGAGTAAGGGAGC |
| **Rect_128nt_G01** | CCAAAATATAATGCAGATACATAAACACCAGAACGAGTAGTGACAAGAACCGGATATACCAAGCGCGAAACATGCCACT ACGAAGGCATGCGCCGACAATGACACTCCAAAAGGAGCCTTACAACGCC |
| **Rect_128nt_G02** | AAATAATTTTAAATTGTAAACGTTGATATTCAACCGTTCTAAATGCAATGCCTGAGAGGTGGCATCAATTCTTTTAGTTTG ACCATTACCAGACCGGAAGCAAAAAAGCGGATTGCATCAGATAAAAA |
| **Rect_128nt_G03** | CCCCGATTTAGAGCTTGACGGGGAAATCAAAAGAATAGCCGCAAGCGGTCCACGCTCCTAATGAGTGAGCTAGTTTCCTG TGTGAAATTTGGGAAGGGCGATCGCACTCCAGCCAGCTTTGCCATCAA |
| **Rect_128nt_H01** | TGAGTTTCGTCACCAGTACAAACTTAATTGTATCGGTTTAGCTTGATACCGATAGTCCAACCTAAAACGAAATGACCCCC AGCGATTATTCATTACCCAAATCACTTGCCCTGACGAGAACGCCAAAA |
| **Rect_128nt_H02** | GGAATTACTCGTTTACCAGACGACAAAAGATTAAGAGGAACGAGCTTCAAAGCGAAGATACATTTCGCAAATGGGGCGC GAGCTGAAATAATGTGTAGGTAAAGAAATCACCATCAATATAATATTTT |
| **Rect_128nt_H03** | GTTAAAATTTTAACCAATAGGAACCCGGCACCGCTTCTGGTCAGGCTGCGCAACTGTGTTATCCGCTCACAATGTAAAGC CTGGGGTGGGTTTGCCCCAGCAGGGGCAAAATCCCTTATAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAACAAACT AT |
| **Rect_192nt_B01** | ACGTTAGTAAATGAATTTTCTGTAAGCGGAGTGAGAATAGCTTTTGCGGGATCGTCGGGTAGCAACGGCTACTTACTTAG CCGGAACGCTGACCAACTTTGAAAAGAACTGGCTCATTATTTAATAAAACGAACTAGCGTCCAATACTGCGGAATGCTTT AAACAGTTGATGGCTTAGAGCTTATTTAAATA |
| **Rect_192nt_B02** | TGCAACTAAGCAATAAAGCCTCAGTTATGACCCTGTAATATTGCCTGAGAGTCTGGAAAACTAGCATGTCAAGATTCTCC GTGGGAACCGTTGGTGTAGATGGGGGGTAACGCCAGGGTTGTGCCAAGCTTGCATGCATTAATGAATCGGCCCGCCAGG GTGGTTTTTAACGTCAAAGGGCGAAGAACCATC |
| **Rect_192nt_C01** | CAAAATTAAAGTACGGTGTCTGGAAGAGGTCATTTTTGCGCAGAAAACGAGAATGAATGTTTAGACTGGATAACGGAAC AACATTATTACCTTATGCGATTTTAGAGGACAGATGAACGGCGCGACCTGCTCCATGAGAGGCTTTGAGGACTAGGGAGT TAAAGGCCGAAAGGAACAACTAAAGCTTTCCAG |
| **Rect_192nt_C02** | ACCCAAATCAAGTTTTTTGGGGTCAAAGAACGTGGACTCCCTTTTCACCAGTGAGACCTGTCGTGCCAGCTGCCTGCAGG TCGACTCTGCAAGGCGATTAAGTTCGCATCGTAACCGTGCGAGTAACAACCCGTCGTCATATGTACCCCGGTAAAGGCTA TCAGGTCACTTTTGCGGGAGAAGCAGAATTAG |
| **Rect_192nt_D01** | CGTAACGATCTAAAGTTTTGTCGTGAATTGCGAATAATAAGGTCGCTGAGGCTTGCAAAGACTTTTTCATGAAAATTGTG TCGAAATCTGTACAGACCAGGCGCTTAATCATTGTGAATTACAGGTAGAAAGATTCAGGGGGTAATAGTAAACCATAAA TCAAAAATCATTGCTCCTTTTGATAAGTTTCAT |
| **Rect_192nt_D02** | TCCATATACATACAGGCAAGGCAACTTTATTTCAACGCAATTTTTGAGAGATCTACTGATAATCAGAAAAGCAACATTAA ATGTGAGCATCTGCCAGTTTGAGGGAAAGGGGGATGTGCTAGAGGATCCCCGGGTACTTTCCAGTCGGGAAACGGGCAA CAGCTGATTACAAGAGTCCACTATTGAGGTGCC |
| **Rect_192nt_E01** | CAATAAATACAGTTGATTCCCAATTTAGAGAGTACCTTTAAGGTCTTTACCCTGACAAAGAAGTTTTGCCAGATCAGTTG AGATTTAGTGGTTTAATTTCAACTATAGGCTGGCTGACCTTGTATCATCGCCTGATGGAAGTTTCCATTAAACATAACCGA TATATTCTTTTTTCACGTTGAAAATAGTTAG |
| **Rect_192nt_E02** | GTAAAGCACTAAATCGGAACCCTAGTTGTTCCAGTTTGGAGCCCTTCACCGCCTGGTTGCGCTCACTGCCCGCCGAGCTC GAATTCGTTATTACGCCAGCTGGCGGACGACGACAGTATCGTAGCCAGCTTTCATCCCCAAAAACAGGAAGACCGGAGA GGGTAGCTAGGATAAAAATTTTTAGTTAACATC |
| **Rect_192nt_F01** | TGTAGCATTCCACAGACAGCCCTCATCTCCAAAAAAAAGGACAACCATCGCCCACGCGGGTAAAATACGTAAAAGTACA ACGGAGATTTCATCAAGAGTAATCTTAAATTGGGCTTGAGAGAATACCACATTCAACGCGAGAGGCTTTTGCATATTATA GTCAGAAGCCTCCAACAGGTCAGGATCTGCGAA |
| **Rect_192nt_F02** | CGAGTAGAACTAATAGTAGTAGCAAACCCTCATATATTTTAGCTGATAAATTAATGTTGTATAAGCAAATATCGCGTCTG GCCTTCCTGGCCTCAGGAAGATCGGTGCGGGCCTCTTCGCAATCATGGTCATAGCTACTCACATTAATTGCGCCCTGAGA GAGTTGCACGAGATAGGGTTGAGTAAGGGAGC |
| **Rect_192nt_G01** | TCAATTCTTTTAGTTTGACCATTACCAGACCGGAAGCAAAAAAGCGGATTGCATCAGATAAAAACCAAAATATAATGCAG ATACATAAACACCAGAACGAGTAGTGACAAGAACCGGATATACCAAGCGCGAAACATGCCACTACGAAGGCATGCGCC GACAATGACACTCCAAAAGGAGCCTTACAACGCC |
| **Rect_192nt_G02** | CCCCGATTTAGAGCTTGACGGGGAAATCAAAAGAATAGCCGCAAGCGGTCCACGCTCCTAATGAGTGAGCTAGTTTCCTG TGTGAAATTTGGGAAGGGCGATCGCACTCCAGCCAGCTTTGCCATCAAAAATAATTTTAAATTGTAAACGTTGATATTCA ACCGTTCTAAATGCAATGCCTGAGAGGTGGCA |
| **Rect_192nt_H01** | TGAGTTTCGTCACCAGTACAAACTTAATTGTATCGGTTTAGCTTGATACCGATAGTCCAACCTAAAACGAAAGAGGCAAA AGAATACACTCCCTCAGAGCCGCCCCACCACCGGAACCGCCTAAAACACTCATCTTTGACCCCCAGCGATTATTCATTAC CCAAATCACTTGCCCTGACGAGAACGCCAAAA |
| **Rect_192nt_H02** | GGAATTACGAGGCATAGTAAGAGCTTTTTAAGAAAAGTAATATCTTACCGAAGCCCAACACTATCATAACCCTCGTTTAC CAGACGACAAAAGATTAAGAGGAACGAGCTTCAAAGCGAAGATACATTTCGCAAATGGTCAATAACCTGTTTAATCAAT AATCGGCTGCGAGCATGTAGAAACCAGCTATAT |

**Table S1. Staple list of the rectangle design.** Grey-colored staples were changed to longer staples.

| Name | Sequence (5'→3') |
|---|---|
| **Staples used for 6HB design** | |
| **6HB_A01** | AATACTTGAGCATATACAGGCAAGGCAACTATATTCGCAAAT |
| **6HB_A02** | CGCAAGGTACATTTTTCATTT |
| **6HB_A03** | AGAATTAGCAATAAAGCCTCATTGCGGGATTTCAA |
| **6HB_A04** | CTGACTAAAGATTAGTACCTTTACTAATAGTAGTACGGATTG |
| **6HB_A05** | CATCCAAGAGCTGAGTTTGACCATTAGAATAAAAAACCCTGT |
| **6HB_A06** | AACGAGTATATATTCAGAAGCAAAAAAACATTATGTTTTTAG |
| **6HB_A07** | GGGGCGCTAAATCAAAGCTAAATCGGTTGTACCAGGCATTAA |
| **6HB_A08** | CATCAAATTATAGTTTAAATGCAATGCCTTCCCAATGCTCCT |
| **6HB_A09** | AACCCTCAGATTTAAAAGGTGGCATCAATTCTAATTTCTGCG |
| **6HB_B01** | TTTGATATTAGAGAAGAGGAAGCCCGAAATCAGGTTGTGTAG |
| **6HB_B02** | GCGTTTTGAGAATGGGGTGAGAAAGGCCGCAACTACTTAATT |
| **6HB_B03** | GTGAAGTTTCAAAAACCATAAATCAAAAAGACTTCCCAACAG |
| **6HB_B04** | GCTCATTTTACCTTTATTCAACCGTTCTGAGGGGGACTAATG |
| **6HB_B05** | CACCATCTTCTCAACATGTTTTAAATATGGAGACAACAGTTC |
| **6HB_B06** | ATAAATAATAATGCTGTAGAGACTGGATAGCGTCATAATAGT |
| **6HB_B07** | AGAAAACAATTCGAGCTTCCATTGAATCCCCCTTAAATCGTC |
| **6HB_B08** | TTGTGAAATACCAGTACCACATTCCAATACTGCGGAGAACTG |
| **6HB_B09** | GTAAAGATTCATTCATTTTTGCGGATGGGCAAACTAAATATC |
| **6HB_B10** | GCTGAATTTAAAGCGAACCAGACCGGAACTTAGAGAAGTACG |
| **6HB_B11** | GTCAGGAAGAGGTCCATATAACAGTTGATGAGTAACTTTACC |
| **6HB_B12** | AAAATGTAATATGAATGCGATTTCAAATGCTTTAAGTCAAAT |
| **6HB_C01** | GCAAAAGAAGGCACGAGAGTCTGGAGCAAATCTTGTCCATGT |
| **6HB_C02** | CAACACTATAACAACATTATTACAGGTAAGGCATAAATAGCG |
| **6HB_C03** | TCAGGTCTTGTTTACCAGACGACGATAAATCTACAGAACGAG |
| **6HB_C04** | TTGAGATATAACGCAAGAAGTTTTGCCAAGCTGATTTAATCA |
| **6HB_C05** | TCAGTGAATCATAACCCTCCATTACCCAAATCAGCACAAGAA |
| **6HB_C06** | TGCCGGATTTGCAACAAAAGGAATTACGGAAAGATTCTACGT |
| **6HB_C07** | CACTACGAATACACCCGCGACCTACGTAACAAAGCGAAAGAG |
| **6HB_C08** | AGAGGCTGAGGGTATGAGATGGTTTAATAGAAAAATCATCAG |
| **6HB_C09** | TAGTAAAACGAACTAACGGAAGGCTTGCCCTGAACTGCTCAT |
| **6HB_C10** | CCGGATAATTGCCTCAACCTAAACGAGAAACACCAAAGGCTA |
| **6HB_C11** | TAATAAATTGGGCTGCTATTTTTGAGAGAAACCAAGTAAGAG |
| **6HB_C12** | CAGATACTTAGGAATCAGGACGTTGGGATTCAACTAAATTAA |
| **6HB_D01** | TGAACGGTGGCTGAGAGGCGCAGACGGTATCATCGCCAGCGA |
| **6HB_D02** | AATTGTGCCGGAACCCTTCATCAAGAGTAACAAGAGTAATGC |
| **6HB_D03** | CATAGGCTAATCGTTCCATTAAACGGGTTTGACCCCCTGATA |
| **6HB_D04** | TACTTAGTCGAAATTAAAACACTCATCTAAAATACGAATCGA |
| **6HB_D05** | GCGGGATCGAAGAGGACAGATGAACGGTAATCATATGTACCC |
| **6HB_D06** | CCGAACTGACCAACTTTGATCACCCTCAGCAGCTGTATCGGT |
| **6HB_D07** | TAAATGAATTTTCTAAAAGCCCCAAAAATAAAGGCCGCTTTT |
| **6HB_D08** | TTATACCAAGCGCGAAACAACGGCTACAGAGGCTTTGCTAAA |
| **6HB_D09** | CGGTTGATAATCAGGTATGGGATTTTGAGGACTAAAGACTTT |
| **6HB_D10** | CAACTTTCAACAGTGCCTTTAATGAAAGACAGCATCGGAACG |
| **6HB_D11** | AGGGTAGCAAAGTACAACGGAGATTTGTCAATCATAAGGGAA |
| **6HB_D12** | TTCATGAGGAAGTTAAAACTAGCATGTCGTACAGACCAGGCG |
| **6HB_E01** | AAACTACGTTGCGCCGACAATGTACCGTAACACCCAGCCCAA |
| **6HB_E02** | AGGCTCCCTTGCTTGGCTTGCAGGGAGTCAGGAAGACGTTAG |
| **6HB_E03** | ACCGATAAAATAATTTTTTCACGTTGAATTAAACAACCGATA |
| **6HB_E04** | GCCCCCTAACAGTGTGTTAAATCAGCTCGATAGCAGTCGAGA |
| **6HB_E05** | TCGCATTCCATGACAACAACCATCGCCCTATTTTGTCATAGT |
| **6HB_E06** | AAGCAAATCGCTGATCGAGGTGAATTTCAATCTCCAGGAACA |
| **6HB_E07** | TAGCGTAGAATTGCGAATACGCCTGTAGCATTCACCCAGTAC |
| **6HB_E08** | CTTGAGTGCCTATTGGATAAGTGTGAGTTTCGTCAAGTTAAT |
| **6HB_E09** | TATTCGGTATTTAAAAAGTTTTGTCGTCAATAGAAAAAAAAA |
| **6HB_E10** | TAGGAACAAATTTTCCCGTATAACACAGACAGCCCTTAAAAT |
| **6HB_E11** | TTATCAGAAAAGGATTCAGCGGAGTGAGTTTCCAGATTGTAT |
| **6HB_E12** | ACTAAAGACGATCTATTGTAAACGTTAAACGCATAGCTTGAT |
| **6HB_F01** | GCTCAGTTATAAGTCCCTCATTTTCAGGATTTTTTTCAGTGC |
| **6HB_F02** | CCTCAGACCATCAATGGTAATAAGTTTTTCTGAAAGGGTTTT |
| **6HB_F03** | AGGAACGGCCACCAATAGCCCGGAATAGATTAGCGCATGAAA |
| **6HB_F04** | GGGTTGAACCAGGCTCGGAACCTATTATAACGGGGAACCAAT |
| **6HB_F05** | GTATTAAGAGGCTGAGACTCGTTCCAGTAAGCGGAATCAAGT |
| **6HB_F06** | TGAATTTACCCTCAAGAGAAGGATTAGGGTGTATCACCGTAC |
| **6HB_F07** | ATACAGGAGTGTACAAATAATTCGCGTCCTCAGAACCGCCAC |

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|---|---|
| **6HB_F08** | CAGCACCGTAATCACATTAAATGTGAGCTTGATATTCACAAA |
| **6HB_F09** | CAAATAAATCACCCTCAGAACCGCCACCTGGCCTTCCTGTAG |
| **6HB_F10** | TTGCCTTTAGCGTCCCACCGGAAGAATGGAAAGCGCAGTCTC |
| **6HB_F11** | CCAGCTTTCATCAAGTAGCGACATCATACATGGCTTTTGATG |
| **6HB_F12** | TCAGGAGGTTTAGTACCGCCCTCATTAAAGCCACCGCCTCCC |
| **6HB_G01** | TTTGGGAAGCCGCCACCAGAAGGTGAATTATCAGCCGGAAAT |
| **6HB_G02** | ATAGGTCTAAACCACCACCAGAGCCGCCTTGACCGCCATTAC |
| **6HB_G03** | CGGATTCAGGCAGGCTCAGAACCGCCACATAATCAGGCATTT |
| **6HB_G04** | CGGAACCCGCCACCTCAGACGATTGGCCGAGTAACTCGATAG |
| **6HB_G05** | TACCAGAAAGTAAGTGTAGATGGGCGCAATATTGAAACATAT |
| **6HB_G06** | CCCTCAGATTAGCGTTTGCCATCTTTTCCCTCAGATTGACAG |
| **6HB_G07** | GAGGTTGTCCGTGGGAAACGTCACCAATTTTCATCAAATCAC |
| **6HB_G08** | TTAAGAAAGGAAACATAAAGGTGCCGTCACCGACTACAAAGT |
| **6HB_G09** | CATTAGCTAGCCCCCTTATTAGAGCCAGCAAAAGATGAGCCA |
| **6HB_G10** | TCGGTCAAAGGCCGGAACAAACGGCGGAGCCAGCAGCCACCA |
| **6HB_G11** | TATTCATACGTTGGCAGATAGCCTCACCAGTAGCATAATGGG |
| **6HB_G12** | TCAGAGCAGAGCCAAGACTGTAGCGCGTGAAACCAAACCCGT |
| **6HB_H01** | ATCGCACCAATGGTTTACCAGCGCCAAATCGGCCTAAGAGCA |
| **6HB_H02** | TAGAAAAACGCAAAGGGAGGGAAGGTAATCGTAACGCCCTTT |
| **6HB_H03** | AGATAACTAGAAAATTCATAAGTCAGAGGGTAAGTCTGAACA |
| **6HB_H04** | CAATCAACCAAGACTCCTTATTACGCAGAGTTTATGGGCGAC |
| **6HB_H05** | CTGCCAGCGATTGAGACACCACGGAATATATGTTAGAATACC |
| **6HB_H06** | ACTTGCGGCGAGGCAGCTTTCCGGCACCGAATTAATACAAAA |
| **6HB_H07** | TAAGAACGGAGGTTAATTTGCCATTGAGCGCTAATCCTCCCG |
| **6HB_H08** | AGAAACAACTGGCATGATTACAAGAATTGAGTTAAATCAGAG |
| **6HB_H09** | ATTCAACTTTGAGGAGCAATAGCTATCTAATAACGGCAAACG |
| **6HB_H10** | CAAAAGAATGAAATGGACGACGACAGTAGACAAAATTTGTCA |
| **6HB_H11** | CCCTGAATCCAGCCGTTTTAGCGAAGCCCAATAATCAGGAAG |
| **6HB_H12** | AAAAGAATACATACCGAGGAAACGCAATTACCGAACGTGCAT |
| **6HB_I01** | TCTTTCCCCATATTCGCATTAGACGGGAGCTTCTGGGTATTC |
| **6HB_I02** | GCAAGCCTTAACGTCAAAATATTAAACCAAGTACGTCATTCC |
| **6HB_I03** | TTTTTGTGTTTTTATCCTGAATCTTACCAAATAAGAATAACA |
| **6HB_I04** | AATTGAGAAGCCAACGGTGCGGGCCTCTTTCCTTAACAATAA |
| **6HB_I05** | AAACCAGAGGGAAGATTTATCCCAATCCAACGCTAAGTTGCT |
| **6HB_I06** | TTGGGAAGGATGAAAATAGCAGCCTTTAAGGCTGCCCGCGCC |
| **6HB_I07** | TAAAAACGCAAAGCTCAGATATAGAAGGCAAGATTACGAGCG |
| **6HB_I08** | CAATAGCACCCAGCTACAATTTTATTTTCATCGGTAGAACAA |
| **6HB_I09** | CAGTATAAATCGCCTCCAGACGACCGCACTCATCGAGGGCTT |
| **6HB_I10** | AAGAACGGGGCGATCGCTCAACATAGGAATCATTAGCAACTG |
| **6HB_I11** | TAAACAGAGAGCCTTTGAAGCCTTAAATCTTATCCGTGCCGG |
| **6HB_I12** | ATTTTGCAAGCAAAGCCATTCGCCATTCCAGAGAGAAACGAT |
| **6HB_J01** | ATAGATATACAGTAATAAGAGAATATAACCTGTTTCGAGCAT |
| **6HB_J02** | AAGGCGTAGTCCTGAACAATTAATGGTTTGAAAAAATCTTCT |
| **6HB_J03** | AGTGAATAACAGTACCCAGTCACGACGTTAATTTCATCATAG |
| **6HB_J04** | GCTGGCGCCAATCATAATGCAGAACGCGAGTACCGTAATTTA |
| **6HB_J05** | AACGCCAATGAAAAATAATATCCCATCCCGATTAATTACTAG |
| **6HB_J06** | GTAAAGTGTTCAGCATAATCGGCTGTCTTCGCTATTTCTTAC |
| **6HB_J07** | GTAGAAAAAAGGGGGTATCATATGCGTTCAACATGACAAAAG |
| **6HB_J08** | TAATGGAAACCTTGAATTTATCATACCGACCGTGTATATGTG |
| **6HB_J09** | AAAAAGCGCATTTTCGAGCAATAAGAATAAACAATGATAAAT |
| **6HB_J10** | GGCAGAGCTGTTTAGATGTGCTGCAAGGTAATTTAATCAACA |
| **6HB_J11** | ACAACATAATTCTGATATTTAACAACGCATACAAATACGCCA |
| **6HB_J12** | GACCTAAGGGTTTTCATAAATCACCGGAATCATAAGTTGGGT |
| **6HB_K01** | CAGTGCCCTTTTTCCTTTTTAACCTCCGATTAAGAAATTAAT |
| **6HB_K02** | CCCGGGTACCTGATGCAAATCCAATCGCGACTCTAAGAAAAC |
| **6HB_K03** | ATATAACGCAAACATAGCGATAGCTTAGGCTTAGGAGAACGC |
| **6HB_K04** | AAGAGTCAGACTACAAATATATTTTAGTTGTAAAACCTTTTT |
| **6HB_K05** | TACCTGATATATGTAAATGAAAATCGCGCAGAGAATTGAATA |
| **6HB_K06** | CAAAGAATGAGTAATCGAATTCGTAATCGATTGCTCCTACCA |
| **6HB_K07** | AAAAGTTACCACCAAAGGGTTAGGCGAATTATTCATGCGGAA |
| **6HB_K08** | AAAATTATAGAATCCTTGAAAAAGAAGATGATGTTTTTCAAT |
| **6HB_K09** | GAGAAAAAAGCTTGTTAACAATTTCATTCGCTATTCGCTGAG |
| **6HB_K10** | CCAAGTTACCGAGCCATTATCATAAACAAACATCAGAGGATC |
| **6HB_K11** | GTCTGAGAATAGTGCTTCTGTAAATCGTTGAATTACGACGGC |
| **6HB_K12** | TTTCCCTATTACATCATGCCTGCAGGTCAAGACAATTGGGTT |
| **6HB_L01** | GTAGATTGATTCATCAATATAATCCTGAAATAAAGCTTTTAC |

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|---|---|
| **6HB_L02** | TACGAGCCTGTCAGATGAATATACAGTAAATTCCAACAATTC |
| **6HB_L03** | TATTAGTGGCACAGATAAAGTGTAAAGCTCTAAAAGTGCCAC |
| **6HB_L04** | TACATTTTTCAGGTTTAACAACAACTAATAGATCATATCTTT |
| **6HB_L05** | TTCCTGTGAAACAAGCACGTAAAACAGATTGTTTGTATTCCT |
| **6HB_L06** | CTTCTGAATTATTTTAACGGATTCGCCTATGGTCATAATTTT |
| **6HB_L07** | AATACGTCTTTAATCGCCTGCAATAGAGCCGTCAAGAATGGC |
| **6HB_L08** | ATCGGGAGTGAAATAATCCTTTGCCCGAATCATCAGATTATA |
| **6HB_L09** | GACAACTAGATGATGGCAAGGATTTAGAAGTATTTTAGATAA |
| **6HB_L10** | GATTATCCGTATTATGTTATCCGCTCACACAGTACAAATTGC |
| **6HB_L11** | TATCAAAATAATGGGAAGGAGCGGAATTACGTTATTAGCTGT |
| **6HB_L12** | AGGAGCACGGAAGCACAATATTTTAGACTTTACAACACAACA |
| **6HB_M01** | GGCAGATCTCAAATATCAAACGCTCAATCGTCTAAAAATACC |
| **6HB_M02** | GTCAGTAGCCAGCAATTGAGGAAGGTTACTGGGGTGCGTAAG |
| **6HB_M03** | CTTTCCATGACCCTCAATCAATATCTGGGCGCTCAGGACATT |
| **6HB_M04** | GCTGAACTCACCACCAGCAGAAGATAAATAAAGCAGCAAATC |
| **6HB_M05** | GGGAGCTGTGCTTTAACCTGTCGTGCCACTCATGGTTAACCG |
| **6HB_M06** | GAGTGAGGAAAGGAGCAAATGAAAAATCACAGAGGACATCGC |
| **6HB_M07** | AACAGTTCTAACTCTAGAACCCTTCTGACCCTAAATGAGGCG |
| **6HB_M08** | CTGGCCAAATACCGAACGAACCAGTCACACGACAATTACATT |
| **6HB_M09** | GTATAACAAACAGGCATCACGCAGAAATGGATTATTCAGAGC |
| **6HB_M10** | CATTAAAACAGAGAACATTAATTGCGTTTCAGTTGTCACCTT |
| **6HB_M11** | GCTGAGATTAACACGCGCGAACTGATAGCCTGAAAGCCTAAT |
| **6HB_M12** | TACATTTGTCGGGACCTCGTTAGCAGTAATAAAAGCTGCCCG |
| **6HB_N01** | GAGTAGACGAGAAGTGTTTTTATAATCAAACATCACAATATT |
| **6HB_N02** | AGCCCGAAAAATCCTTTCACCAGTGAGACGAGAAACCATCAC |
| **6HB_N03** | TCGGCCAGCCATTGCTTTGATTAGTAATGTGAGGCGACAGGA |
| **6HB_N04** | GCGGTCAAGAACTCAAACTAAAGGAGCGGGCGCAAGGAAGGG |
| **6HB_N05** | AGGGTGGCGATCGGCCTTGCTGGTAATAGCGTATTGCTTAAT |
| **6HB_N06** | TAAAAGAAATACTTCAACAGGAAAAACGGCTGCATCGAGCAC |
| **6HB_N07** | ACCGCCAACGCGCGCGCGTACTATGGTTGATTTTACACCGAG |
| **6HB_N08** | GCGCCGCGCCAGAATCCTGCTGCGCGTAACCACCAAAGTGTA |
| **6HB_N09** | AATCGGCGATAGGGCACTACGTGTAGGGCGCTGGCAAAGAAT |
| **6HB_N10** | TTGTAGCGTCTGTCAGGCCGATTAAAGGGCTTTGATAATGAA |
| **6HB_N11** | ACGGTACTACAGGGGGGAGAGGCGGTTTTCCAGAACTTGCCT |
| **6HB_N12** | AAGAAAGTTTTTCTCTTATAAATCACACCCGCCGCGGGCGCC |
| **6HB_O01** | GCTTGACTCACCGCCGAAAATCCTGTTTAGTTTGGCCGTCTA |
| **6HB_O02** | TTGCCCTGGGGAAATTGGGGTCGAGGTGCGAAAAAAACAAGA |
| **6HB_O03** | TCAGGGCAAGTTTTGCCGGCGAACGTGGCGGGCAAGTTCCGA |
| **6HB_O04** | GTCCACTCAGCAGGCTGGCCCTGAGAGAGCCCCCGGCACTAA |
| **6HB_O05** | CCAAATCGATGGCCTTGAGTGTTGTTCCGATGGTGCAGCTGA |
| **6HB_O06** | TTTGCCCATTAAAGCCAACGTCAAAGGGCCGTAAAATTTAGA |
| **6HB_Long_A01** | GTGTCTGTTCAAAAACCATAAATCAAAAAGACTTCCCAACAGGTCAGGAAGAGGTCCATTCCATATAACATGAGTAACT TTACCCTGACTAAAGATTAGTACCTTTACTAATAGTAGTACGGATTG |
| **6HB_Long_A02** | CATCAAATTATAGTTTAAATGCAATGCCGTTGATTCCCAATTCTGCGAAATATATTCAGAAGCAAAAAAACATTATGTTT TTAGAACCCTCCGAGTAGAAAGGTGGCATCAATTCTAATTGCTCCT |
| **6HB_Long_A03** | CGCAAGGTTAGATATTCATTTGGGGCGCTAAATCAAAGCTAAATCGGTTGTACCAGGCATTAACATCCAAGAGCTGAAT TTAGTTTGACCAATAAAAAACCCTGTAATACTTGAGCATATACAGGC |
| **6HB_Long_B01** | AAAATGTTTCTCAACATGTTTTAAATATGGAGACAGTCAAATCACCATCAATATGAATGCGATTTCAAATGCTTTAAAC AGTTCAGAAAACGAGAATGGGGTGAGAAAGGCCGCAACTAAAGTACG |
| **6HB_Long_B02** | TTTGATATTAGAGAAGAGGAAGCCCGAAATCAGGTTGTGTAGGTAAAGAGAAGTTTATTTTTGCGGATGGGCAAACTAA ATATCGCGTTTTAATTCGAGCTTCCATTGAATCCCCCTTAAGAACTG |
| **6HB_Long_B03** | GCTCATTATACCAGTACCACATTCCAATACTGCGGAATCGTCATAAATATTAAAGCGAACCAGACCGGAACTTAGAGCT TAATTGCTGAATATAATGCTGTAGAGACTGGATAGCGTCAACTAATG |
| **6HB_Long_C01** | AGAGGCTGAGGGTATGAGATGGTTTAATAGAAAAATCATCAGTTGAGATATAACGCAAGAAGTTTTGCCAAGCTGATTT AATCATTGTGAATTACCTTTATTCAACCGTTCTGAGGGGGTAATAGT |
| **6HB_Long_C02** | CAGATACTTAGGAATCAGGACGTTGGGATTCAACTAAATTAATGCCGGATTTGCAACAAAAGGAATTACGGAAAGATT CTACGTTAATAAATTGGGCTGCTATTTTTGAGAGAAACCAAGTAAGAG |
| **6HB_Long_C03** | TCAGGTCTTGTTTACCAGACGACGATAAATCTACAGAACGAGTAGTAAAACGAACTAACGGAAGGCTTGCCCTGAACTG CTCATTCAGTGAATAACAACATTATTACAGGTAAGGCATAAATAGCG |
| **6HB_Long_D01** | CAACACTATCATAACCCTCCATTACCCAAATCAGCTCCATGTTACTTAGTCGAAATTAAAACACTCATCTAAAATACGA ATCGATGAACGGTGGCTGAGAGGCGCAGACGGTATCATCGCCAGCGA |
| **6HB_Long_D02** | CACTACGAATACACCCGCGACCTACGTAACAAAGCGAAAGAGGCAAAAGAAGGCACGAGAGTCTGGAGCAAATCTTG ACAAGAACCGGATAATTGCCTCAACCTAAACGAGAAACACCAAAGGCTA |
| **6HB_Long_D03** | TTCATGAGGAAGTTAAAACTAGCATGTCGTACAGACCAGGCGCATAGGCTAATCGTTCCATTAAACGGGTTTGACCCCC TGATAAATTGTGCCGGAACCCTTCATCAAGAGTAACAAGAGTAATGC |
| **6HB_Long_E01** | CCGAACTGACCAACTTTGATCACCCTCAGCAGCTGTATCGGTTTATCAGAAAAGGATTCAGCGGAGTGAGTTTCCAGAT TGTATAAGCAAATCGCTGATCGAGGTGAATTTCAATCTCCAGGAACA |
| **6HB_Long_E02** | TTATACCAAGCGCGAAACAACGGCTACAGAGGCTTTGCTAAACAACTTTCAACAGTGCCTTTAATGAAAGACAGCATCG GAACGAGGGTAGCAAAGTACAACGGAGATTTGTCAATCATAAGGGAA |
| **6HB_Long_E03** | TAAATGAATTTTCTAAAAGCCCCAAAAATAAAGGCCGCTTTTGCGGGATCGAAGAGGACAGATGAACGGTAATCATAT GTACCCCGGTTGATAATCAGGTATGGGATTTTGAGGACTAAAGACTTT |
| **6HB_Long_F01** | TAGGAACAAATTTTCCCGTATAACACAGACAGCCCTTAAAATTCGCATTCCATGACAACAACCATCGCCCTATTTTGTCA |

| | |
|---|---|
| | TAGTTAGCGTAGAATTGCGAATACGCCTGTAGCATTCACCCAGTAC |
| **6HB_Long_F02** | AAACTACAAATAATTTTTTCACGTTGAATTAAACAACCGATATATTCGGTATTTAAAAAGTTTTGTCGTCAATAGAAAAA AAAAAGGCTCCCTTGCTTGGCTTGCAGGGAGTCAGGAAGACGTTAG |
| **6HB_Long_F03** | ACTAAAGACGATCTATTGTAAACGTTAAACGCATAGCTTGATACCGATAGTTGCGCCGACAATGTACCGTAACACCCGT CGAGAGGGTTGAACCAGGCTCGGAACCTATTATAACGGGGAACCAAT |
| **6HB_Long_F04** | GCTCAGTTATAAGTCCCTCATTTTCAGGATTTTTTTCAGTGCCTTGAGTGCCTATTGGATAAGTGTGAGTTTCGTCAAGTT AATGCCCCCTAACAGTGTGTTAAATCAGCTCGATAGCAAGCCCAA |

**Table S2. Staple list of the 6HB design.** Grey-colored staples were changed to longer staples.

| Staples used for mini hexagon structure | |
|---|---|
| **Name** | **Sequence (5'→3')** |
| **MiniHex_105_E01-1** | CTGAGAGTCTGGAATTCAACCGTTCTAGCTGATAGTCATTGC |
| **MiniHex_105_E01-2** | CAAAGGCTATCAGAATTAATGCCGGAGAGGGTAGCTATTTATTCATT |
| **MiniHex_105_E01-3** | CATCAAGAGTAATCAACGTAACAAAGCTGCTCATCTGACCTT |
| **MiniHex_105_E01-4** | ACCCAAATCTTGACAAGAACCGGATTTGAGAGATCTA |
| **MiniHex_105_E02-1** | AGATGAACGGTGTCCTGCTCCATGTTACTTAGCCAAGAGGAC |
| **MiniHex_105_E02-2** | TGACCAACTTTGAGGAACGAGGCGCAGACGGTCAATCATGGAAGGGC |
| **MiniHex_105_E02-3** | CATTCGCCATTCACGGGCCTCTTCGCTATTACGCCAAAGCGC |
| **MiniHex_105_E02-4** | GATCGGTGGGCTGCGCAACTGTTGAAGGGAACCGAAC |
| **MiniHex_105_E03-1** | GCTTTCCGGCACCGATGGGCGCATCGTAACCGTGTCCAGCCA |
| **MiniHex_105_E03-2** | AGGAAGATCGCACCATCTGCCAGTTTGAGGGGACGACGACGGTTGAT |
| **MiniHex_105_E03-3** | TAAAACTAGCATGAAGCCCCAAAAACAGGAAGATGGTAATCG |
| **MiniHex_105_E03-4** | AATCAGAATCAATCATATGTACCCCAGTATCGGCCTC |
| **MiniHex_105_E04-1** | GGATTGACCGTAAGCCAGGGTTTTCCCAGTCACGCAAACGGC |
| **MiniHex_105_E04-2** | TTCTCCGTGGGAAACGTTGTAAAACGACGGCCAGCCGTCGGA |
| **MiniHex_105_E04-3** | AGCGAGTAACAACTGCCAAGCTTGCATGCCTGCATAAATGTG |
| **MiniHex_105_E04-4** | CTTTCATCAACATGGTCGACTCTAGAGGATCCCCGTAGCCAG |
| **MiniHex_105_E05-1** | AACGCCATCAAAACAATTCTGCGAACGAGTAGATCCAATAGG |
| **MiniHex_105_E05-2** | GCTCATTTTTTAATTAGTTTGACCATTAGATACATTAAATCA |
| **MiniHex_105_E05-3** | CATTAAATTTTTGTTTCGCAAATGGTCAATAACCAAAATTCG |
| **MiniHex_105_E05-4** | TTAATATTTTGTTTGTTTAGCTATATTTTCATTTTGTAAACG |
| **MiniHex_105_E06-1** | GTCTGGAAGTTTCTGAAATTGTTATCCGCTCACAAGTACGGT |
| **MiniHex_105_E06-2** | AATATGCAACTAAATTCCACACAACATACGAGCCATGTTTTA |
| **MiniHex_105_E06-3** | GCTGTAGCTCAACGGAAGCATAAAGTGTAAAGCCAATATAAT |
| **MiniHex_105_E06-4** | AGCTTAATTGCTGTGGGGTGCCTAATGAGTGAGCTGGCTTAG |
| **MiniHex_105_E07-1** | AATTGCTCCTTTTAAATATCGCGTTTTAATTCGAGTACCTTT |
| **MiniHex_105_E07-2** | TCAGGATTAGAGAGCTTCAAAGCGAACCAGACCGGAAGCATAAATCA |
| **MiniHex_105_E07-3** | TCTACTAATAGTAAGGCAAAGAATTAGCAAAATTGCATCAAT |
| **MiniHex_105_E07-4** | TACAGGCAGTAGCATTAACATCCAAAACTCCAACAGG |
| **MiniHex_105_E08-1** | CATCAAAAAGATTAAACCTGTCGTGCCAGCTGCAGCGGATTG |
| **MiniHex_105_E08-2** | AGTCAGAAGCAAATTAATGAATCGGCCAACGCGCACTATTAT |
| **MiniHex_105_E08-3** | GGTCTTTACCCTGGGGGAGAGGCGGTTTGCGTATAAAAATCA |
| **MiniHex_105_E08-4** | ATGACCATAAATCTGGGCGCCAGGGTGGTTTTTCAAACGAGA |
| **MiniHex_105_E09-1** | ATTGAATCCCCCTAAAGAAGTTTTGCCAGAGGGGAAATATTC |
| **MiniHex_105_E09-2** | GCGGAATCGTCATGTAATAGTAAAATGTTTAGACTGGATTGTAATAC |
| **MiniHex_105_E09-3** | AATCGGTTGTACCGAGAAGCCTTTATTTCAACGCTAAAGCTA |
| **MiniHex_105_E09-4** | TTTTGCGGAAAAACATTATGACCCAGCGTCCAATACT |
| **MiniHex_105_E10-1** | GACGATAAAAACCCCCTGAGAGAGTTGCAGCAAGTACCAGAC |
| **MiniHex_105_E10-2** | CATAACCCTCGTTCGGTCCACGCTGGTTTGCCCCAACACTAT |
| **MiniHex_105_E10-3** | GCATAGTAAGAGCAGCAGGCGAAAATCCTGTTTGATTACGAG |
| **MiniHex_105_E10-4** | AACGCCAAAAGGAATGGTGGTTCCGAAATCGGCAAGATACAT |
| **MiniHex_105_E11-1** | AGTTGAGATTTAGCAGTCAGGACGTTGGGAAGAAGATTCATC |
| **MiniHex_105_E11-2** | TTACAGGTAGAAAAAATCTACGTTAATAAAACGAACTAAAGGTAAAG |
| **MiniHex_105_E11-3** | ATTTTAAATGCAAGGGTGAGAAAGGCCGGAGACACCTCATAT |
| **MiniHex_105_E11-4** | ATTCAAAATGCCTGAGTAATGTGTCGGAACAACATTA |
| **MiniHex_105_E12-1** | TACCTTATGCGATGTTCCAGTTTGGAACAAGAGTTTGTGAAT |
| **MiniHex_105_E12-2** | TCAACTTTAATCACCACTATTAAAGAACGTGGACGTTTAATT |
| **MiniHex_105_E12-3** | TGGGCTTGAGATGTCCAACGTCAAAGGGCGAAAATAGTAAAT |
| **MiniHex_105_E12-4** | ACACCAGAACGAGACCGTCTATCAAACAAAGTACGACGAGAA |
| **MiniHex_105_V01-1** | AGGGGGATGTTAAGGCGATTAAGTTGGGTAACTGGGATAGGTCTTTTTTCGTTGG |
| **MiniHex_105_V01-2** | TGTAGCTTCTGGTGCTTTTTTTTGGAAACCAGGCAGCTGGCGAA |
| **MiniHex_105_V02-1** | CTGGTCAGTGAATAATTTTTTGGCTTGCCCTAACGGAGATTT |
| **MiniHex_105_V02-2** | GTATCATCGCTAAATTGTGTCGAAATCCGCGAACAGACCAGGCTTTTTTTCATAGG |
| **MiniHex_105_V03-1** | TCGAATTCGTTTGGTCATAGCTGTTTCCTGTGATTCCATATAATTTTTTAGTTGA |
| **MiniHex_105_V03-2** | TTCCATAATTCGCGTTTTTTTTTGGCCTTCCTGGGTACCGAGC |
| **MiniHex_105_V04-1** | AAGAGTTTTTTTATCGATGAACTGTATAAGCAATTTTTTATATTTAAATGGGGC |
| **MiniHex_105_V04-2** | GCGAGCTTTTTTTGAAAAGGTGAAGCAATAAAGTTTTTTCCTCAGAGCAAAGGA |
| **MiniHex_105_V04-3** | TAAAAATTTTTTTTTTTTAGAACGTCAAATCACCTTTTTTCAATATGATGCAAAC |
| **MiniHex_105_V05-1** | AAATCAAAAGTCCCGAGATAGGGTTGAGTGTTTTTAAGAACTGTTTTTTCTCATT |
| **MiniHex_105_V05-2** | ATACGAATACCACATTTTTTTTCAACTAATGCAAATCCCTTAT |
| **MiniHex_105_V06-1** | AATTGCGTTGTACTGCCCGCTTTCCAGTCGGGAAGAGGAAGCCTTTTTTGAAAGA |
| **MiniHex_105_V06-2** | CTTCGATAAGAGGTCTTTTTTTTTTTTTGCGGATAACTCACATT |
| **MiniHex_105_V07-1** | GAGACGGGCATTGATTGCCCTTCACCGCCTGGAAAATAGCGAGTTTTTTGGCTTT |
| **MiniHex_105_V07-2** | TGCACAAATGCTTTATTTTTTTACAGTTCAGATTTTCACCAGT |
| **Minihex_105_Long_E01** | AGGTCATTGCCTGAGAGTCTGGAATTCAACCGTTCTAGCTGATAAATTAATGCCGGAGAGGGTAGCTATTTATT |

| | |
|---|---|
| | CATTACCCAAATCAACGTAACAAAGCTGCTCATCTGACCTTCATCAAGAGTAATCTTGACAAGAACCGGATTT GAGAGATCTACAAAGGCTATC |
| **Minihex_105_Long_E02** | GAAAGAGGACAGATGAACGGTGTCCTGCTCCATGTTACTTAGCCGGAACGAGGCGCAGACGGTCAATCATGG AAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAAAGCGCCATTCGCCATTCAGGCTGCGCAACTGTTGA AGGGAACCGAACTGACCAACTTT |
| **Minihex_105_Long_E03** | ACTCCAGCCAGCTTTCCGGCACCGATGGGCGCATCGTAACCGTGCATCTGCCAGTTTGAGGGGACGACGACGG TTGATAATCAGAAAAGCCCCAAAAACAGGAAGATGGTAATCGTAAAACTAGCATGTCAATCATATGTACCCCA GTATCGGCCTCAGGAAGATCGC |
| **Minihex_105_Long_E04** | AACAAACGGCGGATTGACCGTAAGCCAGGGTTTTCCCAGTCACGACGTTGTAAAACGACGGCCAGTGCCAAG CTTGCATGCCTGCAGGTCGACTCTAGAGGATCCCCGTAGCCAGCTTTCATCAACATTAAATGTGAGCGAGTAA CAACCCGTCGGATTCTCCGTGGG |
| **Minihex_105_Long_E05** | AACCAATAGGAACGCCATCAAAACAATTCTGCGAACGAGTAGATTTAGTTTGACCATTAGATACATTTCGCAA ATGGTCAATAACCTGTTTAGCTATATTTTCATTTTGTAAACGTTAATATTTTGTTAAAATTCGCATTAAATTTTT GTTAAATCAGCTCATTTTTT |
| **Minihex_105_Long_E06** | AAAGTACGGTGTCTGGAAGTTTCTGAAATTGTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCAT AAAGTGTAAAGCCTGGGGTGCCTAATGAGTGAGCTGGCTTAGAGCTTAATTGCTGAATATAATGCTGTAGCTC AACATGTTTTAAATATGCAACT |
| **Minihex_105_Long_E07** | GAGTACCTTTAATTGCTCCTTTTAAATATCGCGTTTTAATTCGAGCTTCAAAGCGAACCAGACCGGAAGCATAA ATCATACAGGCAAGGCAAAGAATTAGCAAAATTGCATCAATTCTACTAATAGTAGTAGCATTAACATCCAAAA CTCCAACAGGTCAGGATTAGA |
| **Minihex_105_Long_E08** | AAGCGGATTGCATCAAAAAGATTAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGA GGCGGTTTGCGTATTGGGCGCCAGGGTGGTTTTTCAAACGAGAATGACCATAAATCAAAAATCAGGTCTTTAC CCTGACTATTATAGTCAGAAGCA |
| **Minihex_105_Long_E09** | ATAAATATTCATTGAATCCCCCTAAAGAAGTTTTGCCAGAGGGGGTAATAGTAAAATGTTTAGACTGGATTGT AATACTTTTGCGGGAGAAGCCTTTATTTCAACGCTAAAGCTAAATCGGTTGTACCAAAAACATTATGACCCAG CGTCCAATACTGCGGAATCGTC |
| **Minihex_105_Long_E10** | TTTACCAGACGACGATAAAAACCCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGCCCCAGCAGGCG AAAATCCTGTTTGATGGTGGTTCCGAAATCGGCAAGATACATAACGCCAAAAGGAATTACGAGGCATAGTAA GAGCAACACTATCATAACCCTCG |
| **Minihex_105_Long_E11** | AAGATTCATCAGTTGAGATTTAGCAGTCAGGACGTTGGGAAGAAAAATCTACGTTAATAAAACGAACTAAAG GTAAAGATTCAAAAGGGTGAGAAAGGCCGGAGACACCTCATATATTTTAAATGCAATGCCTGAGTAATGTGTC GGAACAACATTATTACAGGTAGA |
| **Minihex_105_Long_E12** | CATTGTGAATTACCTTATGCGATGTTCCAGTTTGGAACAAGAGTCCACTATTAAAGAACGTGGACTCCAACGTC AAAGGGCGAAAAACCGTCTATCAAACAAAGTACGACGAGAAACACCAGAACGAGTAGTAAATTGGGCTTGAG ATGGTTTAATTTCAACTTTAAT |
| **Minihex_105_Long_V01** | AGGGGGATGTTAAGGCGATTAAGTTGGGTAACTGGGATAGGTCTTTTTTCGTTGGTGTAGCTTCTGGTGCTTTT TTTGGAAACCAGGCAGCTGGCGAA |
| **Minihex_105_Long_V02** | GTATCATCGCTAAATTGTGTCGAAATCCGCGAACAGACCAGGCTTTTTTTCATAGGCTGGTCAGTGAATAATTT TTTGGCTTGCCCTAACGGAGATTT |
| **Minihex_105_Long_V03** | TCGAATTCGTTTGGTCATAGCTGTTTCCTGTGATTCCATATAATTTTTTAGTTGATTCCATAATTCGCGTTTTTTT TTGGCCTTCCTGGGTACCGAGC |
| **Minihex_105_Long_V04** | AAGAGTTTTTTTATCGATGAACTGTATAAGCAATTTTTTATATTTAAATGGGGCGCGAGCTTTTTTTGAAAAGG TGAAGCAATAAAGTTTTTTCCTCAGAGCAAAGGATAAAAATTTTTTTTTTTTAGAACGTCAAATCACCTTTTTTC AATATGATGCAAAC |
| **Minihex_105_Long_V05** | AAATCAAAAGTCCCGAGATAGGGTTGAGTGTTTTTAAGAACTGTTTTTTCTCATTATACGAATACCACATTTTT TTTCAACTAATGCAAATCCCTTAT |
| **Minihex_105_Long_V06** | AATTGCGTTGTACTGCCCGCTTTCCAGTCGGGAAGAGGAAGCCTTTTTTGAAAGACTTCGATAAGAGGTCTTTT TTTTTTTGCGGATAACTCACATT |
| **Minihex_105_Long_V07** | GAGACGGGCATTGATTGCCCTTCACCGCCTGGAAAATAGCGAGTTTTTTGGCTTTTGCACAAATGCTTTATTTT TTTACAGTTCAGATTTTCACCAGT |

**Table S3. Staple list of the mini hexagon design.** Grey-colored staples were changed to longer staples.

| Staples used for full-sized hexagon structure | |
|---|---|
| **Name** | **Sequence (5' → 3')** |
| **Hex_84_E01-1** | GTTAGAATCTTAATGCGCCGCTACAGGGCACGTGCTTTCCTC |
| **Hex_84_E01-2** | ATGGTTGCTTTCTCCGTGGGAACAAACGGCGGATTATGTGAGCGAGTA |
| **Hex_84_E01-3** | ACAACCCGTCGGATTGACGAGCACGTATAGCGTACT |
| **Hex_84_E02-1** | CTTCCTGTAATTTTTGTTAAATCAGCTCAATTCGCGTCTGGC |
| **Hex_84_E02-2** | ACCAATAGGGCCGGAGAGGGTAGCTATTTTTGAGATCAACCGTTCTAG |
| **Hex_84_E02-3** | CTGATAAATTAATAACGCCATCAAAAATATTTTTTA |
| **Hex_84_E03-1** | GAGACAGTTCATATATTTTAAATGCAATGGTGAGAAAGGCCG |
| **Hex_84_E03-2** | AATGTGTAGACGGTACGCCAGAATCCTGAGAAGTGGCCGATTAAAGGG |
| **Hex_84_E03-3** | ATTTTAGACAGGAGTAAAGATTCAAAAGGCCTGAGT |
| **Hex_84_E04-1** | TCAACGCAAAAGCTAAATCGGTTGTACCAAGAAGCCTTTATT |
| **Hex_84_E04-2** | TATGACCCTGTAATATCCAGAACAATATTACCGCCAAGAACTCAAACT |
| **Hex_84_E04-3** | ATCGGCCTTGCTGGTAATACTTTTGCGGGAAAACAT |
| **Hex_84_E05-1** | GATTAGTACCACTATTAAAGAACGTGGACGCAATACTTCTTT |
| **Hex_84_E05-2** | CCGTTGTATCCAACGTCAAAGGGCGAAAATCACGCAAATTAA |
| **Hex_84_E05-3** | TCTGTCCAACCGTCTATCAGCTAGGGCGCCCGAGTAAAAGAG |
| **Hex_84_E06-1** | GCAAAATTGCATCAATTCTACTAATAGTAAGGCAAAGAATTA |
| **Hex_84_E06-2** | TAACATCCACAATCAATATCTGGTCAGTTGGCAAATTGCTGAACCTCA |
| **Hex_84_E06-3** | AATATCAAACCCTATAAATCATACAGGCAGTAGCAT |
| **Hex_84_E07-1** | AGCAGCAAAGAAGATAAAACAGAGGTGAGCACGCTGAGAGCC |
| **Hex_84_E07-2** | GTATTAACAGCTCAATCGTCTGAAATGGATTATTTGCTCATGGAAATA |
| **Hex_84_E07-3** | CCTACATTTTGACCCGCCTGCAACAGTGCGCGGTCA |
| **Hex_84_E08-1** | GCCATTAATCAAGTTTGCCTTTAGCGTCAGCCCTAAAACATC |
| **Hex_84_E08-2** | AACTGATAGACTGTAGCGCGTTTTCATCGTCTTTAATGCGCG |
| **Hex_84_E08-3** | GCTATTAGGCATTTTCGGTCATAGCCCCCATATTTTTGAATG |
| **Hex_84_E09-1** | CTGAAAGCGGCGAAAATCCTGTTTGATGGGAACCCTTCTGAC |
| **Hex_84_E09-2** | CAGAGATATGGTTCCGAAATCGGCAAAATACATTCTGGCCAA |
| **Hex_84_E09-3** | TAAAAGGGCCCTTATAAATCAAAAGAATAACACGACCAGTAA |
| **Hex_84_E10-1** | ATAGCAGCCCAGCAAAATCACCAGTAGCAAATGAAACCATCG |
| **Hex_84_E10-2** | CATTAGCAACCACCACCAGAGCCGCCGCCAGCATTCACCACCCTCAGA |
| **Hex_84_E10-3** | GCCGCCACCAGAAGGCCGGAAACGTCACCCCATTAC |
| **Hex_84_E11-1** | CGCCACCCCGCCAGGGTGGTTTTTCTTTTCCTCCCTCAGAGC |
| **Hex_84_E11-2** | CGGAACCGCACCAGTGAGACGGGCAACAGCCAGAGCCACCAC |
| **Hex_84_E11-3** | CACCGGAACTGATTGCCCTTCACCGCCTGTCATAATCAAAAT |
| **Hex_84_E12-1** | CAGACGACTAACCGTGCATCTGCCAGTTTAACCCTCGTTTAC |
| **Hex_84_E12-2** | ACTATCATGAGGGGACGACGACAGTATCGTAGTAAGAGCAAC |
| **Hex_84_E12-3** | ACGAGGCAGCCTCAGGAAGATCGCACTCCGCCAAAAGGAATT |
| **Hex_84_E13-1** | TACCACATTAATAAAACGAACTAACGGAATGAGATTTAGGAA |
| **Hex_84_E13-2** | ATTACAGGTTTTAAACAGTTCAGAAAACGAGAATGATATTCATTGAAT |
| **Hex_84_E13-3** | CCCCCTCAAATGCAGAAAGATTCATCAGTCAACATT |
| **Hex_84_E14-1** | ATAGCGTCTAATCAGAAAAGCCCCAAAAAATGTTTAGACTGG |
| **Hex_84_E14-2** | ATAGTAAACAGGAAGATTGTATAAGCAAAGCCAGAGGGGGTA |
| **Hex_84_E14-3** | GAAGTTTTTATTTAAATTGTAAACGTTAAGGCTTTTGCAAAA |
| **Hex_84_E15-1** | TAGCATGTATGCAACTAAAGTACGGTGTCGTAATCGTAAAAC |
| **Hex_84_E15-2** | GATGAACGTGGAAGTTTCATTCCATATAAAAACAAGAGAATC |
| **Hex_84_E15-3** | TCTGGAGCCAGTTGATTCCCAATTCTGCGCATTGCCTGAGAG |
| **Hex_84_E16-1** | TATAATGCGTACCTTTAATTGCTCCTTTTCTTAATTGCTGAA |
| **Hex_84_E16-2** | GGTCATTTTCAGAGGCGAATTATTCATTTCAATTATTTGAATACCAAG |
| **Hex_84_E16-3** | TTACAAAATCGCGTGCGGATGGCTTAGAGGATAAGA |
| **Hex_84_E17-1** | GGAGAAACAAGAAATTGCGTAGATTTTCAACCTTTTACATCG |
| **Hex_84_E17-2** | CGTCAGATGATAACCTGTTTAGCTATATTTTCATTCATTAGATACATT |
| **Hex_84_E17-3** | TCGCAAATGGTCAAATATACAGTAACAGTGGTTTAA |
| **Hex_84_E18-1** | TCAAAATTTCATCAATATAATCCTGATTGAGAACCTACCATA |
| **Hex_84_E18-2** | TATACTTCTAGCACTAACAACTAATAGATTAGAGCGGAAGGTTATCTA |
| **Hex_84_E18-3** | AAATATCTTTAGGGAATAATGGAAGGGTTTTTGGAT |
| **Hex_84_E19-1** | TCATATTCGAAGCGCATTAGACGGGAGAAAGCGGAATTATCA |
| **Hex_84_E19-2** | CCAGAAGGTTAACTGAACACCCTGAACAAAACAAAGAAACCA |
| **Hex_84_E19-3** | TTTTGCGGAGTCAGAGGGTAATTGAGCGCAGTAACATTATCA |
| **Hex_84_E20-1** | TGCCCGAAAGGGAGGGAAGGTAAATATTGGTATTAAATCCTT |
| **Hex_84_E20-2** | GACAACTCACGGAAATTATTCATTAAAGGTTACAAACAATTC |
| **Hex_84_E20-3** | ATTAGACTTGAATTATCACCGTCACCGACAGGATTTAGAAGT |
| **Hex_84_E21-1** | AAAGACAAAAAGACACCACGGAATAAGTTGTTTACCAGCGCC |
| **Hex_84_E21-2** | TCACAATCACAAATAAATCCTCATTAAAGCCAGAAAGACGATTGGCCT |
| **Hex_84_E21-3** | TGATATTCACAAAATAGAAAATTCATATGTATTTTG |
| **Hex_84_E22-1** | CATAAAGGCCAAAAGAACTGGCATGATTATAGAAAATACATA |
| **Hex_84_E22-2** | TTATTACGCGCCCCCTGCCTATTTCGGAACCTATTTAACAGTGCCCGT |

| | |
|---|---|
| **Hex_84_E22-3** | ATAAACAGTTAATAGTATGTTAGCAAACGAGACTCC |
| **Hex_84_E23-1** | AATAAGTTATTAATTGCGTTGCGCTCACTGGAGTGTACTGGT |
| **Hex_84_E23-2** | ATGATACAGCCCGCTTTCCAGTCGGGAAATACATGGCTTTTG |
| **Hex_84_E23-3** | AAGCGTCACCTGTCGTGCCAGCTGCATTATTACCGTTCCAGT |
| **Hex_84_E24-1** | AGTCAGGACCATTCGCCATTCAGGCTGCGGGCTCATTATACC |
| **Hex_84_E24-2** | TAAGAACTCAACTGTTGGGAAGGGCGATCCCTTATGCGATTT |
| **Hex_84_E24-3** | GTGAATTAGGTGCGGGCCTCTTCGCTATTAACTTTAATCATT |
| **Hex_84_E25-1** | AGTAAATTATCAACGTAACAAAGCTGCTCCACCAGAACGAGT |
| **Hex_84_E25-2** | GAATAAGGCGCAAAGCGGATTGCATCAAAAAGATTCTTTACCCTGACT |
| **Hex_84_E25-3** | ATTATAGTCAGAATTGCCCTGACGAGAAAATTCAGT |
| **Hex_84_E26-1** | CTTGACAAGAAAGAGGACAGATGAACGGTCATCAAGAGTAAT |
| **Hex_84_E26-2** | CCAGGCGCATGCGTTATACAAATTCTTACCAGTATCTAGAAAAAGCCT |
| **Hex_84_E26-3** | GTTTAGTATCATATAGGCTGGCTGACCTTGTACAGA |
| **Hex_84_E27-1** | AAATAAGAGTTAATTTCATCTTCTGACCTTAAATAAGGCGTT |
| **Hex_84_E27-2** | ATGGTTTGACAAAGCGAACCAGACCGGAAGCAAACCAAATATCGCGTT |
| **Hex_84_E27-3** | TTAATTCGAGCTTAATACCGACCGTGTGAAAATTTA |
| **Hex_84_E28-1** | ACGCGAGATCCGGCTTAGGTTGGGTTATACGCAAGACAAAGA |
| **Hex_84_E28-2** | ATGTAAATGATTACATTTAACAATTTCATTTGAATAAACAAACATCAA |
| **Hex_84_E28-3** | GAAAACAAAATTACTGATGCAAATCCAATTAACTAT |
| **Hex_84_E29-1** | TCATAGGTATGCAGAACGCGCCTGTTTATGAATTTATCAAAA |
| **Hex_84_E29-2** | TCAATAGTCAACAATAGATAAGTCCTGAAACGCTGAGAAGAG |
| **Hex_84_E29-3** | AGATTAAGCAAGAAAAATAATATCCCATCATAGCGATAGCTT |
| **Hex_84_E30-1** | AATTAATTTCCCAATCCAAATAAGAAACGAATCGTCGCTATT |
| **Hex_84_E30-2** | CTTCTGTAATTTTTTGTTTAACGTCAAAAGTGAATAACCTTG |
| **Hex_84_E30-3** | ATATGTGAATGAAAATAGCAGCCTTTACATACATAAATCAAT |
| **Hex_84_E31-1** | CAGTTACAACCCAGCTACAATTTTATCCTGAGCCTAATTTGC |
| **Hex_84_E31-2** | ACCAACGCTAGAGCAAGAAACAATGAAATAGCAATCAAGAATTGAGTT |
| **Hex_84_E31-3** | AAGCCCAATAATAAACGAGCGTCTTTCCAGAATCTT |
| **Hex_84_E32-1** | GCCTTAAAGCTTATCCGGTATTCTAAGAAGGGAGGTTTTGAA |
| **Hex_84_E32-2** | CGTTTTAGCAAGCCCAATAGGAACCCATGTACCGTCCACCACCCTCAT |
| **Hex_84_E32-3** | TTTCAGGGATAGCGAACCTCCCGACTTGCCGCGAGG |
| **Hex_84_E33-1** | CCGCCACCATATAAGTATAGCCCGGAATAGCCACCCTCAGAA |
| **Hex_84_E33-2** | CACCGTACTACAAAGTTACCAGAAGGAAACCGAGGTTTAAGAAAAGTA |
| **Hex_84_E33-3** | AGCAGATAGCCGACAGGAGGTTTAGTACCGGTGTAT |
| **Hex_84_E34-1** | CCAGGCGGTCATAGCTGTTTCCTGTGTGAGTTTTGCTCAGTA |
| **Hex_84_E34-2** | TTAGCGGGAATTGTTATCCGCTCACAATTAGAAGGATTAGGA |
| **Hex_84_E34-3** | TCCTCAAGCCACACAACATACGAGCCGGAAAGAGGCTGAGAC |
| **Hex_84_E35-1** | AATCATAATGGGTAACGCCAGGGTTTTCCGGCGCAGACGGTC |
| **Hex_84_E35-2** | CGGAACGACAGTCACGACGTTGTAAAACGCATGTTACTTAGC |
| **Hex_84_E35-3** | ACCTGCTCACGGCCAGTGCCAAGCTTTCTGTCGAAATCCGCG |
| **Hex_84_E36-1** | GGAGATTTAAGAATACACTAAAACACTCAAACAAAGTACAAC |
| **Hex_84_E36-2** | CCCCCAGCGACAACGCCAACATGTAATTTAGGCAGGGCTTAATTGAGA |
| **Hex_84_E36-3** | ATCGCCATATTTAATTATACCAAGCGCGATCTTTGA |
| **Hex_84_E37-1** | GAAGGCACCGCACGACTTAAGTGTCCTTAGTAATGCCACTAC |
| **Hex_84_E37-2** | TAAAATACGTGCTGAATTGTCAACCTTATTCCATTAAACGGG |
| **Hex_84_E37-3** | AGGAAGTTGACAATGTCCCGCCAAAATAAAGACTTTTTCATG |
| **Hex_84_E38-1** | AGCAACGGGGAGTTAAAGGCCGCTTTTGCTCGGAACGAGGGT |
| **Hex_84_E38-2** | TCACCCTCAGTAAAGTAATTCTGTCCAGACGACGAAAGAGAATATAAA |
| **Hex_84_E38-3** | GTACCGACAAAAGGCAGCGAAAGACAGCAGGGATCG |
| **Hex_84_E39-1** | ACCGATATGAATTTCTTAAACAGCTTGATTCGCCCACGCATA |
| **Hex_84_E39-2** | GTTGCGCCGTTATCATTCCAAGAACGGGTATTAAAACCAATCAATAAT |
| **Hex_84_E39-3** | CGGCTGTCTTTCCACAATGACAACAACCAACCGATA |
| **Hex_84_E40-1** | TGTATCGGGATACCGACAGTGCGGCCCTGAGGAGCCTTTAAT |
| **Hex_84_E40-2** | GCTCCAAACCATCTGTAAGCAACTCGTCGCTCCAAAAAAAAG |
| **Hex_84_E40-3** | TTGAAAATGTGGGCACGAATATAGGGGCCTAATTTTTTCACG |
| **Hex_84_E41-1** | GAAAGGAAAGTAAATGAATTTTCTGTATGCGGAGTGAGAATA |
| **Hex_84_E41-2** | GCTAAACAACGTAGGAATCATTACCGCGCCCAATAAACAAGCAAGCCG |
| **Hex_84_E41-3** | TTTTTATTTTCATCTTTCAACAGTTTCAGGGATTTT |
| **Hex_84_E42-1** | ATCTAAAGCCTCCTGGTTGGTGTAATGAGAGTTAGCGTAACG |
| **Hex_84_E42-2** | GCCCTCATTAAACAGGGCTTAAGCTACGTCATTCCACAGACA |
| **Hex_84_E42-3** | GCCTGTAGGGTGCTTGTTACCTCGATAAATACAAACTACAAC |
| **Hex_84_V01-1** | GGAGTTTTTATAATCTTTTTTAGTGAGGCCATGGCAAGTGTA |
| **Hex_84_V01-2** | GCGGTCACGCTGTAACCACCACACCCGCCGCGCAGAGCGGGAGTTTTTTTTAAACA |
| **Hex_84_V02-1** | GTAGAGCCATTGCAATTTTTTCAGGAAAAACACATT |
| **Hex_84_V02-2** | GGCAGATTTTTTTTCACCAGTCGCCCGAGATAG |
| **Hex_84_V02-3** | GGTTGAGTGTTGTTCCAGTTTGGAACAAGAGTATAACATCACTTTTTTTTGCCTGA |
| **Hex_84_V03-1** | GACATTATTAGCGTTTTTTTTTGCCATCTTTGCCCTGAGAGA |

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| **Hex_84_V03-2** | GTTGCAGCAATCCACGCTGGTTTGCCCCAGCAGTAAGAATACGTTTTTTTGGCACA |
| **Hex_84_V04-1** | GATAGGTCACTTTGGTGTAGATGGGCGCATCGGATAAAAACCATTTTTTTAATAGC |
| **Hex_84_V04-2** | GCTTTCTTTTTTTTCAACATTAAGACCGTAATGG |
| **Hex_84_V04-3** | GAGATATTTTGTTAATTTTTATTCGCATTAAGCCA |
| **Hex_84_V05-1** | ACCATTTTTTTTAATATGATATGATCTACAAAGTTTTTTGCTATCAGGTAACGA |
| **Hex_84_V05-2** | ATAAAGTTTTTTCTCAGAGCATAGGATAAAAATTTTTTTTTTAGAACCCCAAATC |
| **Hex_84_V05-3** | GTAGATTTTTTTTTAGTTTGACTGGGGCGCGAGTTTTTTGAAAAGGTGAAGCA |
| **Hex_84_V06-1** | AATCTTTTTTTTAAGCATCACCTCAACAGTTGATTTTTTAAGGAATTGACGTCA |
| **Hex_84_V06-2** | AATCAGTTTTTTAGCGACAGAAAAATACCGAACTTTTTTAACCACCAGCATGAAA |
| **Hex_84_V06-3** | ATAGATTTTTTTAATACATTTGTTGAGCCATTTTTTTTGGAATTAGAGACCGT |
| **Hex_84_V07-1** | GAGCGACAGGAGGTTTTTTTTGAGGCAGGTCTGGAA |
| **Hex_84_V07-2** | AGCGCATTTTTTGTCTCTGAATATGAATCGGCC |
| **Hex_84_V07-3** | AACGCGCGGGTAGAGGCGGTTTGCGTATTGGGTCAGAACCGCCTTTTTTTCCCTCA |
| **Hex_84_V08-1** | CGGCACCGCTTTGCCGGAAACCAGGCAAAGCGCGTTGGGAAGATTTTTTAAATCT |
| **Hex_84_V08-2** | ACGTTCAACTAATGCTTTTTTTGATACATAACAGCCAGCTTTC |
| **Hex_84_V09-1** | TGCGGTTTTTTTATCGTCATAAACCATAAATCATTTTTTAAAATCAGGTAAGAG |
| **Hex_84_V09-2** | CTCAACTTTTTTTGTTTTAAATCAATCATATGTTTTTTTCCCCGGTTGACAATAC |
| **Hex_84_V09-3** | GAAGCCTTTTTTCGAAAGACTTTCCAACAGGTCTTTTTGGATTAGAGATGTAG |
| **Hex_84_V10-1** | GGATTTTTTTTTGCCTGATTGCCCTGAGCAAAATTTTTTGAAGATGATGTACCT |
| **Hex_84_V10-2** | TATCAGTTTTTTTGATGGCAATATTTGCACGTATTTTTTAACAGAAATAAATAAC |
| **Hex_84_V10-3** | TTTTTATTTTTTATGGAAACAGGAGAGAATAACTTTTTTAAAAACAGGCTGAT |
| **Hex_84_V11-1** | TAATTTTTTTTTTAAAAGTTTGTAATATCAGAGTTTTTTAGATAACCCAAGCTA |
| **Hex_84_V11-2** | ACATATTTTTTTAAAGAAACGCAAGGGCGACATTTTTTTCAACCGATTGCGTTAT |
| **Hex_84_V11-3** | TCTTACTTTTTTCGAAGCCCTTAAACGCAATAATTTTTAACGGAATACTGGCA |
| **Hex_84_V12-1** | TGAGATTCTGAAACATTTTTTTGAAAGTATTAGCATAAAGTG |
| **Hex_84_V12-2** | TAAAGCCTGGTCTAATGAGTGAGCTAACTCACTTAACGGGGTCTTTTTTTGTGCCT |
| **Hex_84_V13-1** | CGAAAGGGGGTTGTGCTGCAAGGCGATTAAGTGGGAACCGAACTTTTTTGACCAA |
| **Hex_84_V13-2** | TGAGATTTTTTTTGTTTAATTTCACGCCAGCTGG |
| **Hex_84_V13-3** | CTTTGAACCGGATATTTTTTTCATTACCCAAGGGCT |
| **Hex_84_V14-1** | ACCGGTTTTTTTATCATAATTAAAAGCCAACGCTTTTTTTCAACAGTAGAGGCA |
| **Hex_84_V14-2** | AGACTATTTTTTCTTTTTAACCAAACTTTTTCATTTTTTATATATTTTAATAAAC |
| **Hex_84_V14-3** | TTTTCGTTTTTTAGCCAGTAATCAATAAACAACTTTTTTGTTCAGCTACTGAG |
| **Hex_84_V15-1** | TAGAATTTTTTTCCTTGAAAACCTAATTTACGATTTTTTGCATGTAGAACCAAG |
| **Hex_84_V15-2** | ATTAGTTTTTTTGCTATTTTGCAAATAAACAGCTTTTTTATATTATTTATTCCCT |
| **Hex_84_V15-3** | TACCGCTTTTTTACTCATCGAGGCAAGCAAATCTTTTTGATATAGAAGTCAAG |
| **Hex_84_V16-1** | TGAGTTTTTTTTTCGTCACCAGGACGGAGGATC |
| **Hex_84_V16-2** | GTTGCTCAGAACCGCTTTTTTTACCCTCAGAGAACAC |
| **Hex_84_V16-3** | CCCGGGTACCTAGCTCGAATTCGTAATCATGGATAAGTGCCGTTTTTTTGAGAGG |
| **Hex_84_V17-1** | AGGGTGGATGTCTAAGTGGTTGTGAATTCATGCAACCTAAAACTTTTTTAAAGAG |
| **Hex_84_V17-2** | GCAAGTATCATCGCCTTTTTTTGATAAATTGTCAGGAGAAGCC |
| **Hex_84_V18-1** | ATCTATTTACTCTCGCCCTGGAGTGACTCTATTTTATCAGCTTTTTTTTCTTTCG |
| **Hex_84_V18-2** | GAGGCTTTTTTTTTGAGGACTAACCCCGCTTCTA |
| **Hex_84_V18-3** | AGGTATTCGGTCGCTTTTTTTAGGCTTGCAGCTACA |
| **Hex_84_V19-1** | GACGCATTTCTAATCATTTCTCCGAACTCTGATTTTGTCGTCTTTTTTTTCCAGA |
| **Hex_84_V19-2** | CGTTCAACTAAAGGATTTTTTTTTTGCGAATAATTGAATCGGCT |
| **Hex_84_Long_E01** | GTTAGAATCTTAATGCGCCGCTACAGGGCGCGTACTATGGTTGCTTTCTCCGTGGGAACAAACGGCGGATTATGTGA<br>GCGAGTAACAACCCGTCGGATTGACGAGCACGTATAACGTGCTTTCCTC |
| **Hex_84_Long_E02** | CTTCCTGTAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGGCCGGAGAGGGTAGCTATTTTTGAGATCAACCG<br>TTCTAGCTGATAAATTAATAACGCCATCAAAAATAATTCGCGTCTGGC |
| **Hex_84_Long_E03** | GAGACAGTTCATATATTTTAAATGCAATGCCTGAGTAATGTGTAGACGGTACGCCAGAATCCTGAGAAGTGGCCGAT<br>TAAAGGGATTTTAGACAGGAGTAAAGATTCAAAAGGGTGAGAAAGGCCG |
| **Hex_84_Long_E04** | TCAACGCAAAAGCTAAATCGGTTGTACCAAAAACATTATGACCCTGTAATATCCAGAACAATATTACCGCCAAGAAC<br>TCAAACTATCGGCCTTGCTGGTAATACTTTTGCGGGAGAAGCCTTTATT |
| **Hex_84_Long_E05** | GATTAGTACCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTATCAGCTAGGGCGCCCGAGT<br>AAAAGAGTCTGTCCATCACGCAAATTAACCGTTGTAGCAATACTTCTTT |
| **Hex_84_Long_E06** | GCAAAATTGCATCAATTCTACTAATAGTAGTAGCATTAACATCCACAATCAATATCTGGTCAGTTGGCAAATTGCTGA<br>ACCTCAAATATCAAACCCTATAAATCATACAGGCAAGGCAAAGAATTA |
| **Hex_84_Long_E07** | AGCAGCAAAGAAGATAAAACAGAGGTGAGGCGGTCAGTATTAACAGCTCAATCGTCTGAAATGGATTATTTGCTCAT<br>GGAAATACCTACATTTTGACCCGCCTGCAACAGTGCCACGCTGAGAGCC |
| **Hex_84_Long_E08** | GCCATTAATCAAGTTTGCCTTTAGCGTCAGACTGTAGCGCGTTTTCATCGGCATTTTCGGTCATAGCCCCCATATTTTT<br>GAATGGCTATTAGTCTTTAATGCGCGAACTGATAGCCCTAAAACATC |
| **Hex_84_Long_E09** | CTGAAAGCGGCGAAAATCCTGTTTGATGGTGGTTCCGAAATCGGCAAAATCCCTTATAAATCAAAAGAATAACACGA<br>CCAGTAATAAAAGGGACATTCTGGCCAACAGAGATAGAACCCTTCTGAC |
| **Hex_84_Long_E10** | ATAGCAGCCCAGCAAAATCACCAGTAGCACCATTACCATTAGCAACCACCACCAGAGCCGCCGCCAGCATTCACCAC<br>CCTCAGAGCCGCCACCAGAAGGCCGGAAACGTCACCAATGAAACCATCG |
| **Hex_84_Long_E11** | CGCCACCCCGCCAGGGTGGTTTTTCTTTTCACCAGTGAGACGGGCAACAGCTGATTGCCCTTCACCGCCTGTCATAAT<br>CAAAATCACCGGAACCAGAGCCACCACCGGAACCGCCTCCCTCAGAGC |
| **Hex_84_Long_E12** | CAGACGACTAACCGTGCATCTGCCAGTTTGAGGGGACGACGACAGTATCGGCCTCAGGAAGATCGCACTCCGCCAAA<br>AGGAATTACGAGGCATAGTAAGAGCAACACTATCATAACCCTCGTTTAC |
| **Hex_84_Long_E14** | ATAGCGTCTAATCAGAAAAGCCCCAAAAACAGGAAGATTGTATAAGCAAATATTTAAATTGTAAACGTTAAGGCTTT<br>TGCAAAAGAAGTTTTGCCAGAGGGGGTAATAGTAAAATGTTTAGACTGG |
| **Hex_84_Long_E15** | TAGCATGTATGCAACTAAAGTACGGTGTCTGGAAGTTTCATTCCATATAACAGTTGATTCCCAATTCTGCGCATTGCC |

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| | TGAGAGTCTGGAGCAAACAAGAGAATCGATGAACGGTAATCGTAAAAC |
| **Hex_84_Long_E17** | GGAGAAACAAGAAATTGCGTAGATTTTCAGGTTTAACGTCAGATGATAACCTGTTTAGCTATATTTTCATTCATTAGA TACATTTCGCAAATGGTCAAATATACAGTAACAGTACCTTTTACATCG |
| **Hex_84_Long_E18** | TCAAAATTTCATCAATATAATCCTGATTGTTTGGATTATACTTCTAGCACTAACAACTAATAGATTAGAGCGGAAGGT TATCTAAAATATCTTTAGGGAATAATGGAAGGGTTAGAACCTACCATA |
| **Hex_84_Long_E20** | TGCCCGAAAGGGAGGGAAGGTAAATATTGACGGAAATTATTCATTAAAGGTGAATTATCACCGTCACCGACAGGATT TAGAAGTATTAGACTTTACAAACAATTCGACAACTCGTATTAAATCCTT |
| **Hex_84_Long_E21** | AAAGACAAAAAGACACCACGGAATAAGTTTATTTTGTCACAATCACAAATAAATCCTCATTAAAGCCAGAAAGACG ATTGGCCTTGATATTCACAAAATAGAAAATTCATATGGTTTACCAGCGCC |
| **Hex_84_Long_E23** | AATAAGTTATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTATTACCGT TCCAGTAAGCGTCATACATGGCTTTTGATGATACAGGAGTGTACTGGT |
| **Hex_84_Long_V01** | GGAGTTTTTATAATCTTTTTTAGTGAGGCCATGGCAAGTGTAGCGGTCACGCTGTAACCACCACACCCGCCGCGCAGA GCGGGAGTTTTTTTTAAACA |
| **Hex_84_Long_V02** | GTAGAGCCATTGCAATTTTTTCAGGAAAAACACATTGGCAGATTTTTTTTCACCAGTCGCCCGAGATAGGGTTGAGTG TTGTTCCAGTTTGGAACAAGAGTATAACATCACTTTTTTTTGCCTGA |
| **Hex_84_Long_V03** | GACATTATTAGCGTTTTTTTTTGCCATCTTTGCCCTGAGAGAGTTGCAGCAATCCACGCTGGTTTGCCCCAGCAGTAA GAATACGTTTTTTTGGCACA |
| **Hex_84_Long_V04** | GAGATATTTTGTTAATTTTTATTCGCATTAAGCCAGCTTTCTTTTTTTTCAACATTAAGACCGTAATGGGATAGGTCAC TTTGGTGTAGATGGGCGCATCGGATAAAAACCATTTTTTTAATAGC |
| **Hex_84_Long_V05** | GTAGATTTTTTTTTAGTTTGACTGGGGCGCGAGTTTTTTGAAAAGGTGAAGCAATAAAGTTTTTTCTCAGAGCATAGG ATAAAAATTTTTTTTTTAGAACCCCAAATCACCATTTTTTTTAATATGATATGATCTACAAAGTTTTTTGCTATCAGGT AACGA |
| **Hex_84_Long_V06** | ATAGATTTTTTTAATACATTTGTTGAGCCATTTTTTTTGGAATTAGAGACCGTAATCAGTTTTTTAGCGACAGAAAAAT ACCGAACTTTTTTAACCACCAGCATGAAAAATCTTTTTTTTAAGCATCACCTCAACAGTTGATTTTTTAAGGAATTGA CGTCA |
| **Hex_84_Long_V07** | GAGCGACAGGAGGTTTTTTTTGAGGCAGGTCTGGAAAGCGCATTTTTTGTCTCTGAATATGAATCGGCCAACGCGCG GGTAGAGGCGGTTTGCGTATTGGGTCAGAACCGCCTTTTTTTCCCTCA |

**Table S4. Staple list of the full-sized hexagon design.** Grey-colored staples were changed to longer staples.

| Staples used for thick hexagon structure | |
|---|---|
| **Name** | **Sequence (5'→3')** |
| **ThickHex_84_E01_1** | GAAAGGCTGTAGGTTTGACCGTAATGGGGGAACGCAGCTATTTTTGAGAAGAATCACCAT |
| **ThickHex_84_E01_2** | CAATATGATAAGAGGGTCATCAAAAATAATTCGGCGGAAAAGATTCAGTAGTA |
| **ThickHex_84_E01_3** | GCATTATCCGTGGGAACAAACGCGTCTTAAATTAATGCCGGTTCAACCAGGGTGA |
| **ThickHex_84_E02_1** | AGCTATAGCAATAACATAACCGATATATGATAGTTGAGCTCG |
| **ThickHex_84_E02_2** | ACGTTGTAAAAATCATGAACAGCTTGATACCTCGGTCGATTAGCA |
| **ThickHex_84_E02_3** | AATTCGTACGACGGTTGGGGCGCGAGCTGCAAAGACTGAGGC |
| **ThickHex_84_E02_4** | TGACAACAAAGTCTAGAGGATCCCCGCATGCCCCTGTTT |
| **ThickHex_84_E02_5** | TTGCAGGTTTCTTAGTCATAGCTGTTTCCT |
| **ThickHex_84_E02_6** | AAATTAATTTTCATCCAGTGCCAAGCTTGGGTACCGCGCCGACAA |
| **ThickHex_84_E03_1** | ATAACGGATAAAGACAATAAACAACAAATCATACTGATGC |
| **ThickHex_84_E03_2** | ACCTTTTTAAGAAAACTTGATAAATAAGGCGGCCTGTTATACTTT |
| **ThickHex_84_E03_3** | AAATCCATATAACTTCCCAATTCTGCGAGAAGCCTCTAATGC |
| **ThickHex_84_E03_4** | ACACCGGTGTTCAGTTATTTCAACGCAAAGTTGATATATGTAAAAATGCTG |
| **ThickHex_84_E03_5** | AGAACGCTTAAATAGACAAAGAACGCGACCTCCGGGATTTAG |
| **ThickHex_84_E03_6** | TGCGGGAACGAGTACTTAGGTTGGGTTAATCGCAAAGAATAA |
| **ThickHex_84_E04_1** | CATTCCAATTTAATGATCCCATATACAGGAGTGTACAGGCTGACCTTGAAAACATAGCG |
| **ThickHex_84_E04_2** | TGTCTTTATGTAGACGTCATACATGGCTAGGATTATCTATCATTTTCCCTAAAGAACGGG |
| **ThickHex_84_E04_3** | TATTAAACCAAAAACCGGGATTAGCGGGGTTCAGTAAGAACCAATCAGCCATA |
| **ThickHex_84_E04_4** | CAAGAGATTTGATGCCTAATTTACGAGCCCTTAT |
| **ThickHex_84_E04_5** | TTTAACTGAATTTACCGTTCTTGCTCAACGTCAAAGGGCGAAGTACCGAATCGGC |
| **ThickHex_84_E05_1** | TTAGTAAAGGTAAGATAACCCACAAGTTCCAGAGCTTATC |
| **ThickHex_84_E05_2** | TCGTAGGAATGCGAACCTTTATCCTGAATCTAAGAGCATCGAGCC |
| **ThickHex_84_E05_3** | GAAGCTAACGAGCGTCTAATTGAGATAAAGT |
| **ThickHex_84_E05_4** | CGGTATTGCAAATCGCGTTATACAAATTAGAGAATTTAAGCC |
| **ThickHex_84_E05_5** | CAATAATTACCAACCGCGAGGCGTTTTACATTACCGTATAAA |
| **ThickHex_84_E05_6** | AGTAATACTTACCAGCGCCCAATAGCAACTAA |
| **ThickHex_84_E06_1** | CGCAATTAGTTGCTACCTTTTTCCTTGATATTCACAGTGTATCGGGTTGAGTGTTGTTC |
| **ThickHex_84_E06_2** | GAAACCGGATAGCCTGAGGCAGGTCAGAGTTTAGTTCAAAAGAATAGCCATAATAACGGA |
| **ThickHex_84_E06_3** | ATACCCAAAATTATAAAACCGCCACCCTCAGAGGAGGTGAACAAAGTCAGCCT |
| **ThickHex_84_E06_4** | TCAGGAGCGATTGGAAGAAAAGTAAGCAAGGAAA |
| **ThickHex_84_E06_5** | TTACAGCCGCCAGCATTGACAACCGCCATCGGCAAAATCCCGAACTGGCCAGAAG |
| **ThickHex_84_E07_1** | AAATAAGAGGGTCCATCTTTTCATAACACCGTCAGACACC |
| **ThickHex_84_E07_2** | AACGTAGAAAATAGAAAATATTGACGGAAATCCACCACGACGGGA |
| **ThickHex_84_E07_3** | TTATAAAGGTGAATTATTCAAAATCCCTGAA |
| **ThickHex_84_E07_4** | ACGGAATCATATAAATATTATTTATCCCCTGAACACACCGGA |
| **ThickHex_84_E07_5** | ACCAGAGTATTCATTTTTGTCACAATCAATACATAATAAGAA |
| **ThickHex_84_E07_6** | GAATTAAAATCCAACATAAAGGTGGCAAAAGT |
| **ThickHex_84_E08_1** | ATAAGGGCGGAACGTGATAAATTGTGTCTAACACTGGCCCTGAGAGAGTACCAACTTTGA |
| **ThickHex_84_E08_2** | AAGAGGACAGACCGCCTGAGTTTCGTCACCACATCGCCAGGCGCAGAGATAGC |
| **ThickHex_84_E08_3** | TGTACCGGAAATCCGCGACCACCCTCAGCCAATAGAGCGGTCCACGCTGGT |
| **ThickHex_84_E08_4** | GTTACTTAGCAACCGAACTGACATTCAACCGGCCTGCTCCAT |
| **ThickHex_84_E08_5** | AGCACCACGGAGATTTGTATGTACAAAGCTGATTGCCCTTCATGAACGGTCAATC |
| **ThickHex_84_E09_1** | GAATTTCGGTCAAGAGGCTTTTGCAAGTCAGGAAACAAAG |
| **ThickHex_84_E09_2** | TCAAGAGTAATGACGAGCCTTATGCGATTTTTAATAGTTAGCGTC |
| **ThickHex_84_E09_3** | GTGGGCTCATTATACCAAAGAAGTTTTCATC |
| **ThickHex_84_E09_4** | CTGCTCACATTACCGCAAAATCACCAGTAGCGCGTTTTGCCA |
| **ThickHex_84_E09_5** | GAGGGGGAAGAACTAATAAGGCTTGCCCTCTTGACTTACCAT |
| **ThickHex_84_E09_6** | AGACTGTAGCACCAAAGAACCGGATATTTTCA |
| **ThickHex_84_E10_1** | TGACCATAATTTCAATCGTCATACACTAAAACACTCAGTTTTGGCGGTTTGCGTATTGG |
| **ThickHex_84_E10_2** | CAGAAAATCCCCCTAAACGAAAGAGGCAGTTAGTATCGGCCAACGCGCGTTTAAATCAAA |
| **ThickHex_84_E10_3** | AATCAGGTCTTAATGAAAATGAATTTTCTGTCAACCTACAAATGCTTCATTCA |
| **ThickHex_84_E10_4** | TCCAGACAAAGAATAAATATTCATTGAACGAGAA |
| **ThickHex_84_E10_5** | ACTAATCACTACGAAGGCACATGGGATCGTGCCAGCTGCATTTACCCTAACAGTT |
| **ThickHex_84_E11_1** | CGGAACACACTATCGACGACAGTATCGGACGTTAAACGGTAA |
| **ThickHex_84_E11_2** | AAAGATTAAGACTAGCATATTTAAATTGTAACCTCAGGCATAGTA |
| **ThickHex_84_E11_3** | TCGTAAAAGGAAGCTACAGGTAGAAAGATACGAGGAAGATCGCACT |
| **ThickHex_84_E11_4** | TTAAAATAGGGGACATAACCCTCGTTTAATAAAACGTTTTAA |
| **ThickHex_84_E11_5** | CAGTTTGTCGCATTAGCAAACAAGAGAAATATCGCGAACTAA |
| **ThickHex_84_E11_6** | AGAGCAAACATTATCCGAAAGACTTCAATCGATGATATTTTG |
| **ThickHex_84_E12_1** | GCGATCCCAAAAACAGGTGCCGGAGGACTAAAGACTAAGGAATAGCTAACTCACATTAA |
| **ThickHex_84_E12_2** | ACTGTTGCGCCATTAGCAACGGCTACAGTTTTCACGCCTGGGGTGCCTACCGGTGCGGGC |
| **ThickHex_84_E12_3** | ATCAGAAAAGATGAGTGTGCGAATAATAATTAGGCTTTGAAACCAGGCAAAGGGAAGG |
| **ThickHex_84_E12_4** | CTCTTCGCTAGTGTAAAGTTGAAAATCTCCACGAGGGTCGCCATTCATTAAAT |
| **ThickHex_84_E12_5** | GTGAGCAGACAGCATCGGAAAAAAAAAGCCGGAAGCATAAATTACGCCCTGCGCA |

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|---|---|
| **ThickHex_84_V01_1** | ACTAATAAAGTTCTAGCTGGTAACGCCAGGGTTTTTTTTTTTCCCAGTCACG |
| **ThickHex_84_V01_2** | CCTGTATTTTTTCCAGCTTTCATCAACAGGAGCTGGCGAAAGGGCATACGAGGCTCCA |
| **ThickHex_84_V01_3** | CAATATTTTTTTAATCATACAGGCAAGGAAAAGGTGGTTTTTATCAATTCT |
| **ThickHex_84_V01_4** | GTGTGAAATTTCCGCTCTTGTATCGGTTTTTTCAGCTTGCTT |
| **ThickHex_84_V01_5** | TCGAGGTGAAGAGTTAAAGGCCGCTTTTGATCGTCACCCTCAGCCTTTAAACAATTC |
| **ThickHex_84_V01_6** | CACACAAGGATGTGCTGTTTTTTTAAGGCGATTAAGTTGGAGGCCTT |
| **ThickHex_84_V01_7** | AAAGGAGCAGCGAAGAGTAACAACTTTTTTTCGTCGGATTCACATC |
| **ThickHex_84_V02_1** | CCCCTGCCAGTGCCATCGGTTGTATTTTTTTCCAAAAACATTATGAATTAG |
| **ThickHex_84_V02_2** | GAGCATATCAATAATGCAGGTCGTTATCAAAATCATAGTTTTTTTGTCTGAGAGACT |
| **ThickHex_84_V02_3** | ATACATTTTTTTCGCAAATGGAAGCTAACGTATAAACCCATCGCCCACGAGCCTCA |
| **ThickHex_84_V02_4** | TTTGACCCCCTGTATATCAACAATAGATAAGTCCTGTTTTTTTAACAAGAA |
| **ThickHex_84_V02_5** | ATATATTTTTTTTTTTAGTTAATTTCATCTTCTGACCTTAGAATGACTCCT |
| **ThickHex_84_V02_6** | ATAGCTTAGAGACGCTGGGAACCTATTTTTTCTGAAACATGA |
| **ThickHex_84_V02_7** | AAGTATTAAGTGGTAATAAGTTTTAAC |
| **ThickHex_84_V02_8** | GGAGTGCCTTGAGTAACTATTTCAGAAGAGTCAATAGTGAATACTTAATGC |
| **ThickHex_84_V02_9** | AAATAATGTTTGAAATATTTTTTCCGACCGTGTTTTCAA |
| **ThickHex_84_V03_1** | TTGAGAATCATCACTCATCGAGAACGACTCCAGTACCAGGCGGATACAGTCTCAACGC |
| **ThickHex_84_V03_2** | GAACGTGAAGCAAGCCGTTTTTTTTTTTTATTTTCA |
| **ThickHex_84_V03_3** | CAACATTTTTTTTGTAATTTAGGCAGACTCAACAGTATTTTTGGCTTAA |
| **ThickHex_84_V03_4** | GCCAACGGGCATTTAGAAACAATGAAATAGCAATAGTTTTTTTCTATCTTA |
| **ThickHex_84_V03_5** | TTGCGGGATTTTTTTGGTTTTGAAGCCTTAAATCAAGCGAGATAACCGTAC |
| **ThickHex_84_V03_6** | CAGTTTGGAAAGTCCACTCGAGAGGGTTTTTTATAAGTATAG |
| **ThickHex_84_V03_7** | CCCGGAATAGAACAAATAAATCCTCATTACAGAATGGAAAGCGAGTGCCGTATTAAA |
| **ThickHex_84_V03_8** | CCGAAGCTTTTGCACCCTTTTTTAGCTACAATTCCCGAC |
| **ThickHex_84_V04_1** | TGAAAATAGTACATGATTAAGACTCTTCCGAAACCCTCAGAACCGCAGAGCCGAGAGA |
| **ThickHex_84_V04_2** | ATGGTGGCTTATTACGCTTTTTTTAGTATGTTAGCA |
| **ThickHex_84_V04_3** | ATAACTTTTTTTATAAAAACAGGGAAGTTTGTTTAACTTTTTTCAAAAA |
| **ThickHex_84_V04_4** | ACGATTTCGCATTACGGAACCGCCTCCCTCAGAGCCTTTTTTTGCCACCCT |
| **ThickHex_84_V04_5** | TGGTTTACTTTTTTTCAGCGCCAAAGACAAAAGGGCGTGCAGCAGAACCCA |
| **ThickHex_84_V04_6** | TTGCCCCAGCGAAAATCAGAGCCACCTTTTTTCATTTTCAGG |
| **ThickHex_84_V04_7** | GATAGCAAGCAGCCACCACCCTCAGAGCCCCAGAACCACCACCCACCCTCCTGTTTG |
| **ThickHex_84_V04_8** | CAGAACCATTGAGGGAGTTTTTTGGAAGGTAAATTCATA |
| **ThickHex_84_V05_1** | GAAACCATCCGGTGTACAGACCAGGGGCAACACTACAACGCCTGTAAAGTACAGTAAT |
| **ThickHex_84_V05_2** | TGAGACGCGCATAGGCTTTTTTTTGGCTGACCTTCA |
| **ThickHex_84_V05_3** | CAGTATTTTTTTGCGACAGAATCAAGTGCCGGAAACGTTTTTCACCAAT |
| **ThickHex_84_V05_4** | TAGCAAGTTGCCTTAAAATGTTTAGACTGGATAGCGTTTTTTTTCCAATAC |
| **ThickHex_84_V05_5** | AGAACGAGTTTTTTTTAGTAAATTGGGCTTGAGATGGGGGAGAGTCGTCTT |
| **ThickHex_84_V05_6** | GCGCCAGGGTTTTCTTTACAGACAGCTTTTTATAGTTAGCGT |
| **ThickHex_84_V05_7** | AACGATCTAAATCTTTGACCCCCAGCGATCCAAGCGCGAAACAGCATTCCTCACCAG |
| **ThickHex_84_V05_8** | TGCGGAACTTTAATCATTTTTTTTGTGAATTAAAACACC |
| **ThickHex_84_V06_1** | GGAATACCATAGACTATTATAGTCAAACCTGTTTTGCTAAACAACTGTAATGCGCAGA |
| **ThickHex_84_V06_2** | GTCGGGAGAAGCAAAGCTTTTTTTGGATTGCATCAA |
| **ThickHex_84_V06_3** | TACATTTTTTTTAACGCCAAAAGGAATTTCATCAGTTTTTTTAGATTTA |
| **ThickHex_84_V06_4** | CCAGCCAGCTTTTTTTTTTCCGGCACCGCTTCTGGAAG |
| **ThickHex_84_V06_5** | ATTGTTTTTTTATAAGCAAATGTCAATCATATGTATTTTTTTCCCCGGTTGATA |
| **ThickHex_84_V06_6** | TTGCGTTGCGCTGCCCGGTTTCAGCGTTTTTGAGAATAGAAA |
| **ThickHex_84_V06_7** | GGAACAACTATTTTCATGAGGAAGTTTCCAACGGGTAAAATACTTCAACACTTTCCA |
| **ThickHex_84_V07_01** | CAGTCATACGGTGTCAAATGCAACGTTGGTGTAGATGGGCGCATTTTTTTCGTAAC |
| **ThickHex_84_V07_02** | ATCAGGTCTTTTTTATTGCCTGAGTTCAAAGCGATTTTTTTCCAGACCG |
| **ThickHex_84_V07_03** | ACCAATAATAGGTCATGCCTGAGTAATGCGGAGA |
| **ThickHex_84_V07_04** | TAGCTCAATTTTTTTCATGTTTTAAATATGCAACTAAGATCTACTTTTTTA |
| **ThickHex_84_V07_05** | CGACAATTTTTAGATTTTTTTACCCTCATATATTTTTGGAAGTTTCTTTTTTTCCAT |
| **ThickHex_84_V07_06** | GCTTAGAGTTTTTTTTTAATTGCTGAATATTGATTACTAGAAAATTTTTTGCCTGT |
| **ThickHex_84_V07_07** | ACCGACAATCATATAGATATAGAGCGGATG |
| **ThickHex_84_V07_08** | TCCTTTTGTTTTTTTTAAGAGGTCATTTTTAGGCCTAATTTGCCTTTTTTGTTACA |
| **ThickHex_84_V07_09** | TTTGAATTGAGCGCTTTTTTTAATATCAGAGAAGTAATTCTTTTTTTTTCCAGA |

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|---|---|
| **ThickHex_84_V07_10** | CAAAGTCAACAGCCAAGAAACGCTAATTGC |
| **ThickHex_84_V07_11** | CAGGTCAGTTTTTTTATTAGAGAGTACCTTAAACCGACTTGAGCTTTTTTATTTGG |
| **ThickHex_84_V07_12** | GAAGCAAGTCGTTGGGAAGAATTTTTTAATCTACGTTACCAGACGCATCTGC |
| **ThickHex_84_V07_13** | GGCATTTAGAGCCACAAATCAACACTCCAA |
| **ThickHex_84_V07_14** | TTCGAGCAGTCTGGAAATTTTTGTTATTTTTTTATCAGCTCAAAAGGCT |
| **ThickHex_84_V07_15** | CGTGACGATAAAAATTTTTTTCAAAATAGCGTAGCCCCCTTTTTTTTTTTAGCG |
| **ThickHex_84_Long_E01_1** | GAAAGGCTGTAGGTTTGACCGTAATGGGGGAACGCAGCTATTTTTGAGAAGAATCACCATCAATATGATAA GAGGGTCATCAAAAATAATTCGGCGGAAAAGATTCAGTAGTAGCATTATCCGTGGGAACAAACGCGTCTTA AATTAATGCCGGTTCAACCAGGGTGA |
| **ThickHex_84_Long_E02_1** | AGCTATAGCAATAACATAACCGATATATGATAGTTGAGCTCGAATTCGTACGACGGTTGGGGCGCGAGCTG CAAAGACTGAGGCTTGCAGGTTTCTTAGTCATAGCTGTTTCCT |
| **ThickHex_84_Long_E02_2** | ACGTTGTAAAAATCATGAACAGCTTGATACCTCGGTCGATTAGCAAAATTAATTTTCATCCAGTGCCAAGCT TGGGTACCGCGCCGACAATGACAACAAAGTCTAGAGGATCCCCGCATGCCCCTGTTT |
| **ThickHex_84_Long_E03_1** | ATAACGGATAAAGACAATAAACAACAAATCATACTGATGCAAATCCATATAACTTCCCAATTCTGCGAGAA GCCTCTAATGCAGAACGCTTAAATAGACAAAGAACGCGACCTCCGGGATTTAG |
| **ThickHex_84_Long_E03_2** | ACCTTTTTAAGAAAACTTGATAAATAAGGCGGCCTGTTATACTTTTGCGGGAACGAGTACTTAGGTTGGGTT AATCGCAAAGAATAAACACCGGTGTTCAGTTATTTCAACGCAAAGTTGATATATGTAAAAATGCTG |
| **ThickHex_84_Long_E04_1** | TATTAAACCAAAAACCGGGATTAGCGGGGTTCAGTAAGAACCAATCAGCCATATTTAACTGAATTTACCGTT CTTGCTCAACGTCAAAGGGCGAAGTACCGAATCGGCTGTCTTTATGTAGACGTCATACATGGCTAGGATTAT CTATCATTTTCCCTAAAGAACGGG |
| **ThickHex_84_Long_E04_2** | CAAGAGATTTGATGCCTAATTTACGAGCCCTTATCATTCCAATTTAATGATCCCATATACAGGAGTGTACAG GCTGACCTTGAAAACATAGCG |
| **ThickHex_84_Long_E05_1** | TTAGTAAAGGTAAGATAACCCACAAGTTCCAGAGCTTATCCGGTATTGCAAATCGCGTTATACAAATTAGAG AATTTAAGCCCAATAATTACCAACCGCGAGGCGTTTTACATTACCGTATAAA |
| **ThickHex_84_Long_E05_2** | TCGTAGGAATGCGAACCTTTATCCTGAATCTAAGAGCATCGAGCCAGTAATACTTACCAGCGCCCAATAGCA ACTAAGAAGCTAACGAGCGTCTAATTGAGATAAAGT |
| **ThickHex_84_Long_E06_1** | ATACCCAAAATTATAAAACCGCCACCCTCAGAGGAGGTGAACAAAGTCAGCCTTTACAGCCGCCAGCATTG ACAACCGCCATCGGCAAAATCCCGAACTGGCCAGAAGGAAACCGGATAGCCTGAGGCAGGTCAGAGTTTA GTTCAAAAGAATAGCCATAATAACGGA |
| **ThickHex_84_Long_E06_2** | TCAGGAGCGATTGGAAGAAAAGTAAGCAAGGAAACGCAATTAGTTGCTACCTTTTTCCTTGATATTCACAGT GTATCGGGTTGAGTGTTGTTC |
| **ThickHex_84_Long_E07_1** | AAATAAGAGGGTCCATCTTTTCATAACACCGTCAGACACCACGGAATCATATAAATATTATTTATCCCCTGA ACACACCGGAACCAGAGTATTCATTTTTGTCACAATCAATACATAATAAGAA |
| **ThickHex_84_Long_E07_2** | AACGTAGAAAATAGAAAATATTGACGGAAATCCACCACGACGGGAGAATTAAAATCCAACATAAAGGTGG CAAAAGTTTATAAAGGTGAATTATTCAAAATCCCTGAA |
| **ThickHex_84_Long_E08_1** | AACCGAACTGACATTCAACCGGCCTGCTCCATGTTACTTAGCCGGAACGTGATAAATTGTGTCGAAATCCGC GACCACCCTCAGCCAATAGAGCGGTCCACGCTGGT |
| **ThickHex_84_Long_E08_2** | TGTACCGTAACACTGGCCCTGAGAGAGTACCAACTTTGAAAGAGGACAGACCGCCTGAGTTTCGTCACCAC ATCGCCAGGCGCAGAGATAGCAGCACCACGGAGATTTGTATGTACAAAGCTGATTGCCCTTCATGAACGGT CAATCATAAGGG |
| **ThickHex_84_Long_E09_1** | GAATTTCGGTCAAGAGGCTTTTGCAAGTCAGGAAACAAAGCTGCTCACATTACCGCAAAATCACCAGTAGC GCGTTTTGCCAGAGGGGGAAGAACTAATAAGGCTTGCCCTCTTGACTTACCAT |
| **ThickHex_84_Long_E09_2** | TCAAGAGTAATGACGAGCCTTATGCGATTTTTAATAGTTAGCGTCAGACTGTAGCACCAAAGAACCGGATAT TTTCAGTGGGCTCATTATACCAAAGAAGTTTTCATC |
| **ThickHex_84_Long_E10_1** | AATCAGGTCTTAATGAAAATGAATTTTCTGTCAACCTACAAATGCTTCATTCAACTAATCACTACGAAGGCA CATGGGATCGTGCCAGCTGCATTTACCCTAACAGTTCAGAAAATCCCCCTAAACGAAAGAGGCAGTTAGTA TCGGCCAACGCGCGTTTAAATCAAA |
| **ThickHex_84_Long_E10_2** | TCCAGACAAAGAATAAATATTCATTGAACGAGAATGACCATAATTTCAATCGTCATACACTAAAACACTCA GTTTTGGCGGTTTGCGTATTGG |
| **ThickHex_84_Long_E11_1** | AAAGATTAAGACTAGCATATTTAAATTGTAACCTCAGGCATAGTAAGAGCAAACATTATCCGAAAGACTTC AATCGATGATATTTTGTTAAAATAGGGGACATAACCCTCGTTTAATAAAACGTTTTAA |
| **ThickHex_84_Long_E11_2** | CAGTTTGTCGCATTAGCAAACAAGAGAAATATCGCGAACTAACGGAACACACTATCGACGACAGTATCGGA CGTTAAACGGTAATCGTAAAAGGAAGCTACAGGTAGAAAGATACGAGGAAGATCGCACT |
| **ThickHex_84_Long_E12_1** | ATCAGAAAAGATGAGTGTGCGAATAATAATTAGGCTTTGAAACCAGGCAAAGGGAAGGGCGATCCCAAAA ACAGGTGCCGGAGGACTAAAGACTAAGGAATAGCTAACTCACATTAA |
| **ThickHex_84_Long_E12_2** | CTCTTCGCTAGTGTAAAGTTGAAAATCTCCACGAGGGTCGCCATTCATTAAATGTGAGCAGACAGCATCGGA AAAAAAAAGCCGGAAGCATAAATTACGCCCTGCGCAACTGTTGCGCCATTAGCAACGGCTACAGTTTTCAC GCCTGGGGTGCCTACCGGTGCGGGC |
| | |
| **ThickHex_84_Long_V01_1** | CAATATTTTTTTAATCATACAGGCAAGGAAAAGGTGGTTTTTATCAATTCTACTAATAAAGTTCTAGCTGGT AACGCCAGGGTTTTTTTTTTTCCCAGTCACG |
| **ThickHex_84_Long_V01_2** | GTGTGAAATTTCCGCTCTTGTATCGGTTTTTTCAGCTTGCTTTCGAGGTGAAGAGTTAAAGGCCGCTTTTGAT CGTCACCCTCAGCCTTTAAACAATTC |
| **ThickHex_84_Long_V01_3** | CACACAAGGATGTGCTGTTTTTTTAAGGCGATTAAGTTGGAGGCCTTCCTGTATTTTTTCCAGCTTTCATCAA CAGGAGCTGGCGAAAGGGCATACGAGGCTCCAAAAGGAGCAGCGAAGAGTAACAACTTTTTTTCGTCGGAT TCACATC |
| **ThickHex_84_Long_V02_1** | CCCCTGCCAGTGCCATCGGTTGTATTTTTTTCCAAAAACATTATGAATTAGATACATTTTTTTCGCAAATGGA AGCTAACGTATAAACCCATCGCCCACGAGCCTCAGAGCATATCAATAATGCAGGTCGTTATCAAAATCATA GTTTTTTTGTCTGAGAGACT |
| **ThickHex_84_Long_V02_2** | TTTGACCCCCTGTATATCAACAATAGATAAGTCCTGTTTTTTTAACAAGAAAAATAATGTTTGAAATATTTTT TCCGACCGTGTTTTCAAATATATTTTTTTTTTTTAGTTAATTTCATCTTCTGACCTTAGAATGACTCCT |
| **ThickHex_84_Long_V02_3** | ATAGCTTAGAGACGCTGGGAACCTATTTTTTCTGAAACATGAAAGTATTAAGTGGTAATAAGTTTTAACGGA GTGCCTTGAGTAACTATTTCAGAAGAGTCAATAGTGAATACTTAATGC |
| **ThickHex_84_Long_V03_1** | TTGAGAATCATCACTCATCGAGAACGACTCCAGTACCAGGCGGATACAGTCTCAACGCCAACATTTTTTTTG TAATTTAGGCAGACTCAACAGTATTTTTGGCTTAA |
| **ThickHex_84_Long_V03_2** | GCCAACGGGCATTTAGAAACAATGAAATAGCAATAGTTTTTTTCTATCTTACCGAAGCTTTTGCACCCTTTTT TAGCTACAATTCCCGACTTGCGGGATTTTTTTGGTTTTGAAGCCTTAAATCAAGCGAGATAACCGTAC |
| **ThickHex_84_Long_V03_3** | CAGTTTGGAAAGTCCACTCGAGAGGGTTTTTTATAAGTATAGCCCGGAATAGAACAAATAAATCCTCATTAC AGAATGGAAAGCGAGTGCCGTATTAAAGAACGTGAAGCAAGCCGTTTTTTTTTTTTTATTTTCA |

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| **ThickHex_84_Long_V04_1** | TGAAAATAGTACATGATTAAGACTCTTCCGAAACCCTCAGAACCGCAGAGCCGAGAGAATAACTTTTTTTAT AAAAACAGGGAAGTTTGTTTAACTTTTTTCAAAAA |
| **ThickHex_84_Long_V04_2** | ACGATTTCGCATTACGGAACCGCCTCCCTCAGAGCCTTTTTTTGCCACCCTCAGAACCATTGAGGGAGTTTTT TGGAAGGTAAATTCATATGGTTTACTTTTTTTCAGCGCCAAAGACAAAAGGGCGTGCAGCAGAACCCA |
| **ThickHex_84_Long_V04_3** | TTGCCCCAGCGAAAATCAGAGCCACCTTTTTTCATTTTCAGGGATAGCAAGCAGCCACCACCCTCAGAGCCC CAGAACCACCACCCACCCTCCTGTTTGATGGTGGCTTATTACGCTTTTTTTAGTATGTTAGCA |
| **ThickHex_84_Long_V05_1** | GAAACCATCCGGTGTACAGACCAGGGGCAACACTACAACGCCTGTAAAGTACAGTAATCAGTATTTTTTTGC GACAGAATCAAGTGCCGGAAACGTTTTTCACCAAT |
| **ThickHex_84_Long_V05_2** | TAGCAAGTTGCCTTAAAATGTTTAGACTGGATAGCGTTTTTTTTCCAATACTGCGGAACTTTAATCATTTTTT TTGTGAATTAAAACACCAGAACGAGTTTTTTTTAGTAAATTGGGCTTGAGATGGGGGAGAGTCGTCTT |
| **ThickHex_84_Long_V05_3** | GCGCCAGGGTTTTCTTTACAGACAGCTTTTTATAGTTAGCGTAACGATCTAAATCTTTGACCCCCAGCGATCC AAGCGCGAAACAGCATTCCTCACCAGTGAGACGCGCATAGGCTTTTTTTTGGCTGACCTTCA |
| **ThickHex_84_Long_V06_1** | GGAATACCATAGACTATTATAGTCAAACCTGTTTTGCTAAACAACTGTAATGCGCAGATACATTTTTTTTAA CGCCAAAAGGAATTTCATCAGTTTTTTTAGATTTA |
| **ThickHex_84_Long_V06_2** | CCAGCCAGCTTTTTTTTTTCCGGCACCGCTTCTGGAAGATTGTTTTTTTATAAGCAAATGTCAATCATATGTA TTTTTTTCCCCGGTTGATA |
| **ThickHex_84_Long_V06_3** | TTGCGTTGCGCTGCCCGGTTTCAGCGTTTTTGAGAATAGAAAGGAACAACTATTTTCATGAGGAAGTTTCCA ACGGGTAAAATACTTCAACACTTTCCAGTCGGGAGAAGCAAAGCTTTTTTTGGATTGCATCAA |
| **ThickHex_84_Long_V07_1** | CAGTCATACGGTGTCAAATGCAACGTTGGTGTAGATGGGCGCATTTTTTTCGTAACCGTGACGATAAAAATT TTTTTCAAAATAGCGTAGCCCCCTTTTTTTTTTTAGCG |
| **ThickHex_84_Long_V07_2** | TAGCTCAATTTTTTTCATGTTTTAAATATGCAACTAAGATCTACTTTTTTAACCAATAATAGGTCATGCCTGA GTAATGCGGAGA |
| **ThickHex_84_Long_V07_3** | ACCGACAATCATATAGATATAGAGCGGATGGCTTAGAGTTTTTTTTTAATTGCTGAATATTGATTACTAGAA AATTTTTTGCCTGT |
| **ThickHex_84_Long_V07_4** | TTTGAATTGAGCGCTTTTTTTAATATCAGAGAAGTAATTCTTTTTTTTTCCAGACGACAATTTTTAGATTTTTT TACCCTCATATATTTTTGGAAGTTTCTTTTTTTCCAT |
| **ThickHex_84_Long_V07_5** | CAAAGTCAACAGCCAAGAAACGCTAATTGCTCCTTTTGTTTTTTTTAAGAGGTCATTTTTAGGCCTAATTTGC CTTTTTTGTTACA |
| **ThickHex_84_Long_V07_6** | GGCATTTAGAGCCACAAATCAACACTCCAACAGGTCAGTTTTTTTATTAGAGAGTACCTTAAACCGACTTGA GCTTTTTTATTTGG |
| **ThickHex_84_Long_V07_7** | TTCGAGCAGTCTGGAAATTTTTGTTATTTTTTTATCAGCTCAAAAGGCTATCAGGTCTTTTTTATTGCCTGAGT TCAAAGCGATTTTTTTCCAGACCGGAAGCAAGTCGTTGGGAAGAATTTTTTAATCTACGTTACCAGACGCAT CTGC |

**Table S5. Staple list of the thick hexagon design.** Grey-colored staples were changed to longer staples.

| Linear templates for RCA-produced staples | |
|---|---|
| **Name** | **Sequence (5'→3')** |
| **Template_Rot96_B01** | TTGGTCAGCGTTCCGGCTAAGTAAGTAGCCGTTGCTACCCGACGATCCCGCATCCGCGCTTCGCGCGGATGC GCAAAAGCTATTCTCACTCCGCTTACAGAAAATTCATTTACTAACGT |
| **Template_Rot96_B02** | TATTTAAATAAGCTCTAAGCCATCAACTGTTTAAAGCATTCCGCAGTAAACATCCGCGCTTCGCGCGGATGT TGGACGCTAGTTCGTTTTATTAAATAATGAGCCAGTTCTTTTCAAAG |
| **Template_Rot96_B03** | CACCAACGGTTCCCACGGAGAATCTTGACATGCTAGTTTTCCAGACTCGACATCCGCGCTTCGCGCGGATGT CAGGCAATATTACAGGGTCATAACTGAGGCTTTATTGCTTAGTTGCA |
| **Template_Rot96_B04** | GATGGTTCTTCGCCCTTTGACGTTAAAAACCACCCTGGCGGGCCGATTTGCATCCGCGCTTCGCGCGGATGC ATTAATGCATGCAAGCTTGGCACAACCCTGGCGTTACCCCCCATCTA |
| **Template_Rot96_C01** | CTGGAAAGCTTTAGTTGTTCCTTTCGGCCTTTAACTCCCTAGTCCTCATTCATCCGCGCTTCGCGCGGATGAA GCCTCTCATGGAGCAGGTCGCGCCGTTCATCTGTCCTCTAAAATCG |
| **Template_Rot96_C02** | CATAAGGTAATAATGTTGTTCCGTTATCCAGTCTAAACATTCATTCTCACCATCCGCGCTTCGCGCGGATGGT TTTCTGCGCAAAAATGACCTCTTCCAGACACCGTACTTTAATTTTG |
| **Template_Rot96_C03** | CTAATTCTGCTTCTCCCGCAAAAGTGACCTGATAGCCTTTACCGGGGGTGTCATCCGCGCTTCGCGCGGATGA CATATGACGACGGGTTGTTACTCGCACGGTTACGATGCGAACTTAAT |
| **Template_Rot96_C04** | CGCCTTGCAGAGTCGACCTGCAGGCAGCTGGCACGACAGGTCTCACTGACCATCCGCGCTTCGCGCGGATG GTGAAAAGGGAGTCCACGTTCTTTGACCCCAAAAAACTTGATTTGGGT |
| **Template_Rot96_D01** | TCTGTACAGATTTCGACACAATTTTCATGAAAAAGTCTTTGCAAGCCTTGCATCCGCGCTTCGCGCGGATGC AGCGACCTTATTATTCGCAATTCACGACAAAACTTTAGATCGTTACG |
| **Template_Rot96_D02** | ATGAAACTTATCAAAAGGAGCAATGATTTTTGATTTATGGTTTACTATTACATCCGCGCTTCGCGCGGATGT ACCCCCTGAATCTTTCTACCTGTAATTCACAATGATTAAGCGCCTGG |
| **Template_Rot96_D03** | TGGCAGATGCTCACATTTAATGTTGCTTTTCTGATTATCAGTAGATCTAGCATCCGCGCTTCGCGCGGATGCT CAAAAATTGCGTTGAAATAAAGTTGCCTTGCCTGTATGTATATGGA |
| **Template_Rot96_D04** | GGCACCTCAATAGTGGACTCTTGTAATCAGCTGTTGCCCGTTTCCCGAAGCATCCGCGCTTCGCGCGGATGC TGGAAAGTACCCGGGGATCCTCTAGCACATCCCCCTTTCCCTCAAAC |
| **Template_Rot96_E01** | CTAACTATTTTCAACGTGAAAAAAGAATATATCGGTTATGTTTAATGGTTCATCCGCGCTTCGCGCGGATGA AACTTCCATCAGGCGATGATACAAGGTCAGCCAGCCTATAGTTGAAA |
| **Template_Rot96_E02** | TTAAACCACTAAATCTCAACTGATCTGGCAAAACTTCTTTGTCAGGGTTTCATCCGCGCTTCGCGCGGATGA AAGACCTTAAAGGTACTCTCTAAATTGGGAATCAACTGTATTTATTG |
| **Template_Rot96_E03** | GATGTTAACTAAAAATTTTTATCCTAGCTACCCTCTCCGGTCTTCCTGAACATCCGCGCTTCGCGCGGATGTT TTTGGGGATGAAAGCTGGCTACGATACTGTCGTCGTCCGCCAGCTG |
| **Template_Rot96_E04** | GCGTAATAACGAATTCGAGCTCGGCGGGCAGTGAGCGCAACCAGGCGGCACATCCGCGCTTCGCGCGGATG TGAAGGGCTCCAAACTGGAACAACTAGGGTTCCGATTTAGTGCTTTAC |
| **Template_Rot96_F01** | CTTGATGAAATCTCCGTTGTACTTTTACGTATTTTACCCGCGTGGGCGATCATCCGCGCTTCGCGCGGATGAT GGTTGTCCTTTTTTTTGGAGATGAGGGCTGTCTGTGGAATGCTACA |
| **Template_Rot96_F02** | TTCGCAGATCCTGACCTGTTGGAGGCTTCTGACTATAATATGCAAAAGGGCATCCGCGCTTCGCGCGGATGC CTCTCGCGTTGAATGTGGTATTCTCTCAAGCCCAATTTAAGATTACT |
| **Template_Rot96_F03** | CTGAGGCCAGGAAGGCCAGACGCGATATTTGCTTATACAACATTAATTTACATCCGCGCTTCGCGCGGATGT ATCAGCTAAAATATATGAGGGTTTGCTACTACTATTAGTTCTACTCG |
| **Template_Rot96_F04** | GCTCCCTTACTCAACCCTATCTCGTGCAACTCTCTCAGGGCGCAATTAATCATCCGCGCTTCGCGCGGATGAT GTGAGTAGCTATGACCATGATTGCGAAGAGGCCCGCACCGATCTTC |
| **Template_Rot96_G01** | GGCGTTGTAAGGCTCCTTTTGGAGTGTCATTGTCGGCGCATGCCTTCGTACATCCGCGCTTCGCGCGGATGT AGTGGCATGTTTCGCGCTTGGTATATCCGGTTCTTGTCACTACTCGT |
| **Template_Rot96_G02** | TCTGGTGTTTATGTATCTGCATTATATTTTGGTTTTTATCTGATGCAAGACATCCGCGCTTCGCGCGGATGTC CGCTTTTTTGCTTCCGGTCTGGTAATGGTCAAACTAAAAGAATTGA |
| **Template_Rot96_G03** | TGCCACCTCTCAGGCATTGCATTTAGAACGGTTGAATATCAACGTTTATGCATCCGCGCTTCGCGCGGATGC AATTTAAAATTATTTTTGATGGCAAAGCTGGCTGGAGTGCGATCGCC |
| **Template_Rot96_G04** | CTTCCCAAATTTCACACAGGAAACTAGCTCACTCATTAGGAGCGTGGAGGCATCCGCGCTTCGCGCGGATGC CGCTTGCGGCTATTCTTTTGATTTCCCCGTCAAGCTCTAAATCGGGG |
| **Template_Rot96_H01** | GTAATGAATAATCGCTGGGGGTCATTTCGTTTTAGGTTGGACTATCGGTACATCCGCGCTTCGCGCGGATGT ATCAAGCTAAACCGATACAATTAAGTTTGTACTGGTGACGAAACTCA |
| **Template_Rot96_H02** | AATGTATCTTCGCTTTGAAGCTCGTTCCTCTTAATCTTTTGTCGTCTGACCATCCGCGCTTCGCGCGGATGGT AAACGAGTAATTCCTTTTGGCGTTCTCGTCAGGGCAAGTGATTTGG |
| **Template_Rot96_H03** | GGTGCCGGGTTCCTATTGGTTAAAATTTTAACAAAATATTATATTGATCCCATCCGCGCTTCGCGCGGATGG GTGATTTCTTTACCTACACATTATTTCAGCTCGCGCCCCATTTGCGA |
| **Template_Rot96_H04** | GCCGGCTTTATAAGGGATTTTGCCCCTGCTGGGGCAAACCCACCCCAGGCCATCCGCGCTTCGCGCGGATGG CTTTACATTGTGAGCGGATAACACAGTTGCGCAGCCTGACCAGAAGC |

**Table S6. List of the template strands for RCA process.**